

Manuscript Results

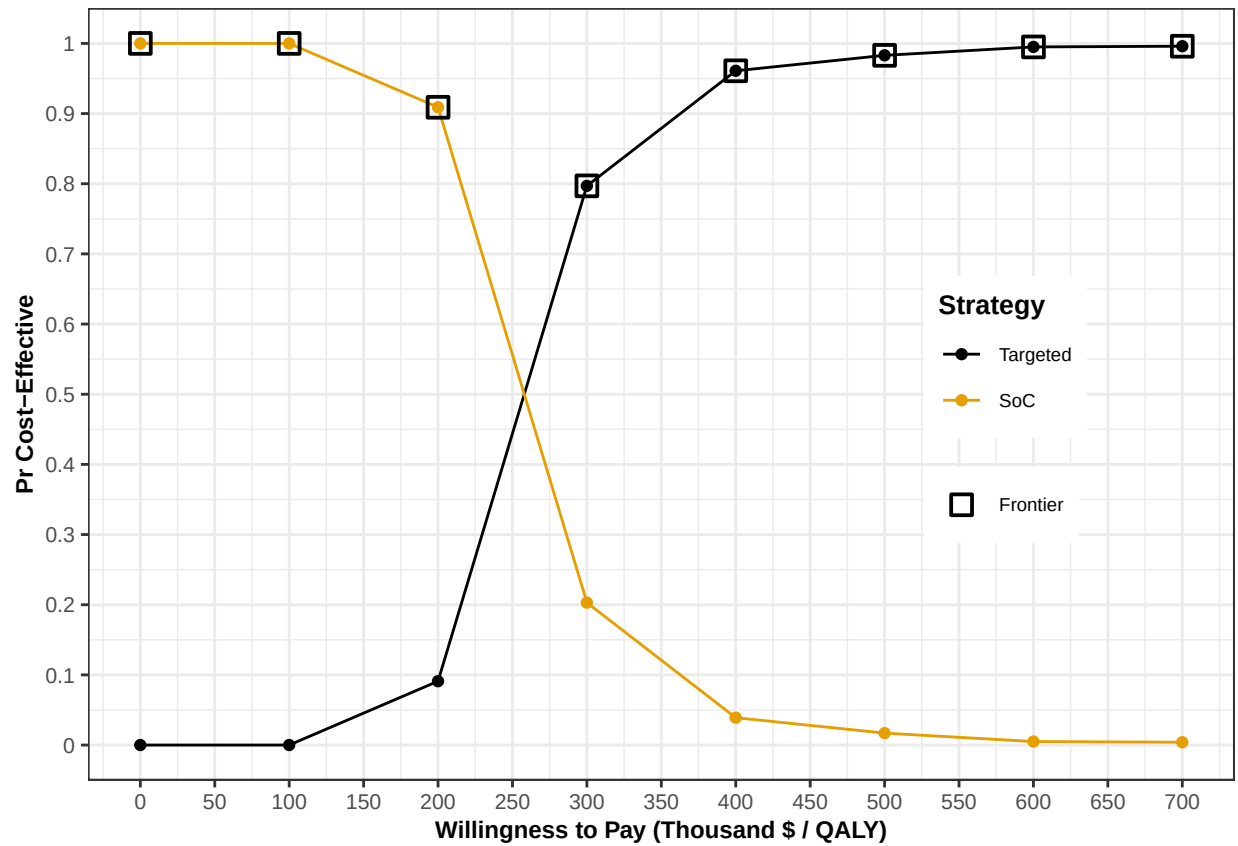
Base case

Scenario #1a

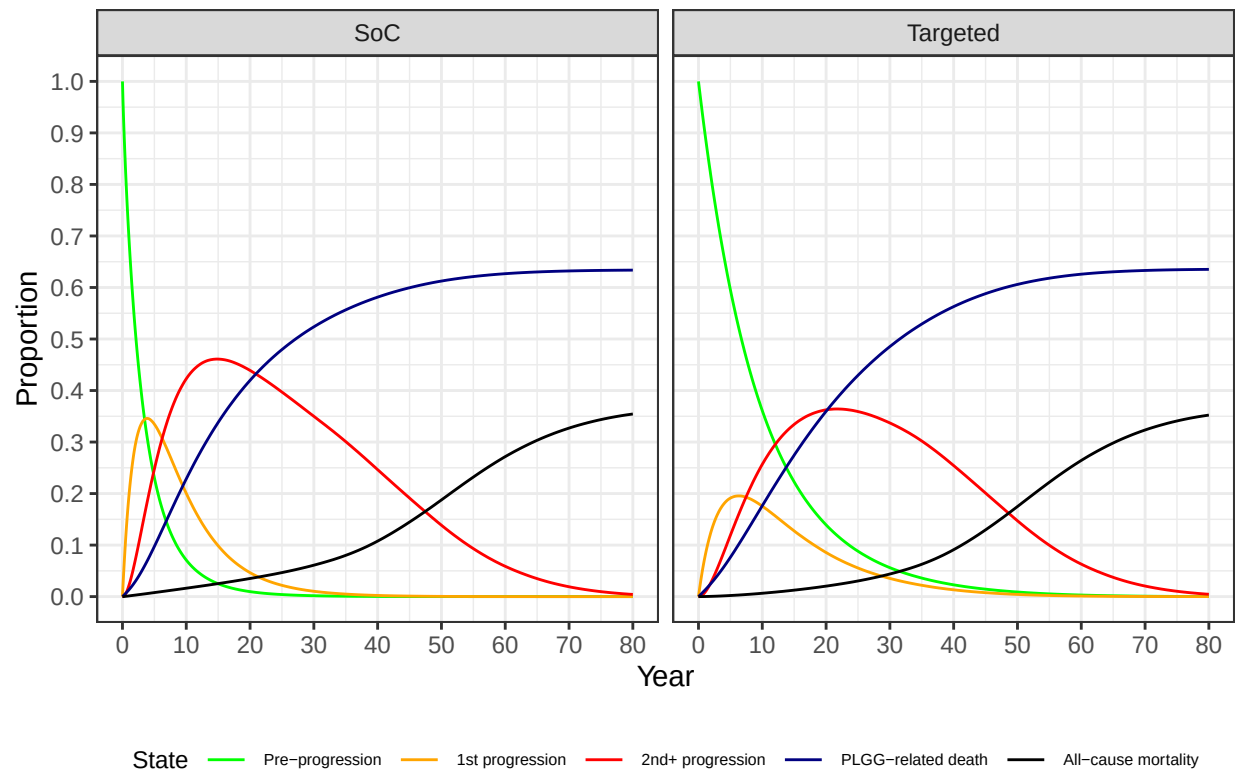
- Combo vs. chemo as first-line treatment
- Lifetime duration of combo therapy
- 0% radiation at progression (Post-2014 SickKids data indicates that no patients who received chemo and progressed were radiated)
- 100% combo price
- Discount rate: 1.5
- Data source for controls & exposed: Investigator assessment - Bouffet et al., NEJM 2023.
- **Post-progression risk for subsequent progression = pre-progression risk for first progression in combo arm
- 10,000 individuals
- 1000 simulation runs

Investigator assessment, Targeted curve – Exponential, SoC curve – Exponential

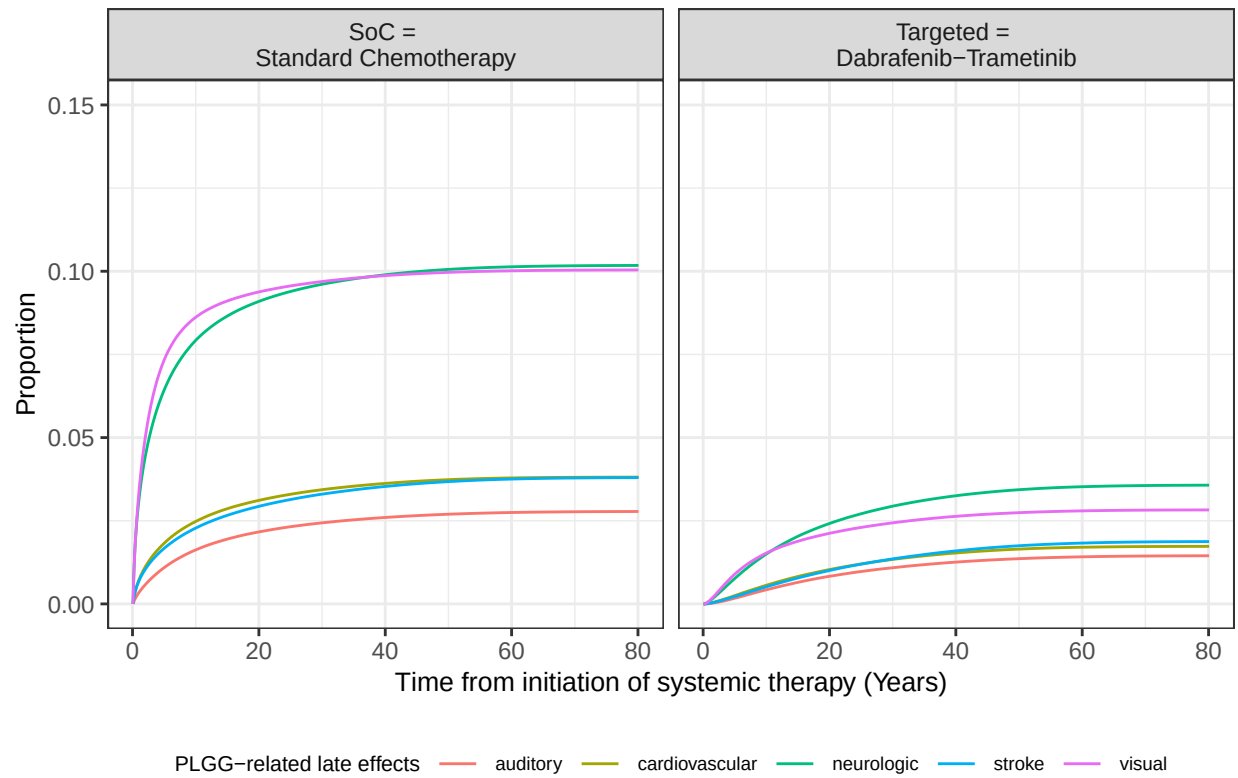
Variable	intervention	control	Delta
Life-years	31.21 (22.69;40.6)	28.62 (19.25;39.7)	2.59 (0.09;7.46)
QALY	14.48 (10.6;18.72)	12.28 (8.51;16.83)	2.21 (0.94;4.36)
Total Cost	\$970,334 (\$720,994;\$1,361,516)	\$415,564 (\$386,410;\$449,808)	\$554,769 (\$311,466;\$942,256)
Non-Targeted PLGG related costs	\$199,724 (\$141,200;\$258,960)	\$273,899 (\$210,008;\$334,620)	\$-74,174 (\$-108,739;\$-47,431)
Late effect costs	\$80,883 (\$29,349;\$145,978)	\$109,507 (\$47,829;\$189,429)	\$-28,623 (\$-51,273;\$-9,141)
General population costs	\$48,922 (\$28,683;\$80,997)	\$32,158 (\$15,260;\$60,585)	\$16,764 (\$9,770;\$28,065)
Targeted costs	\$640,803 (\$376,259;\$1,043,104)	\$0 (\$0;\$0)	\$640,803 (\$376,259;\$1,043,104)
Radiation costs	\$0 (\$0;\$0)	\$0 (\$0;\$0)	\$0 (\$0;\$0)

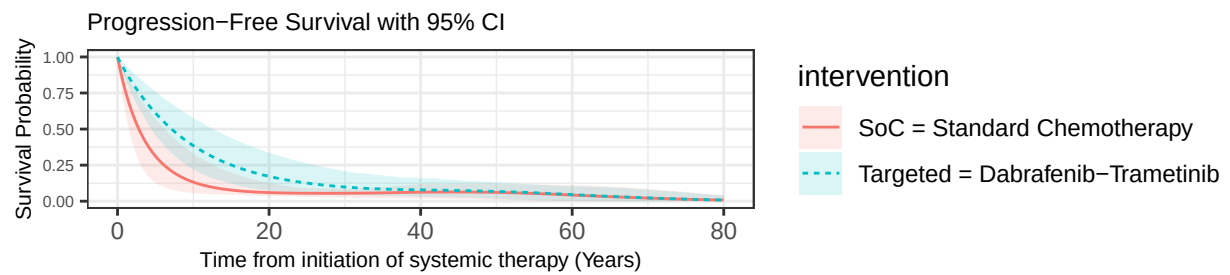
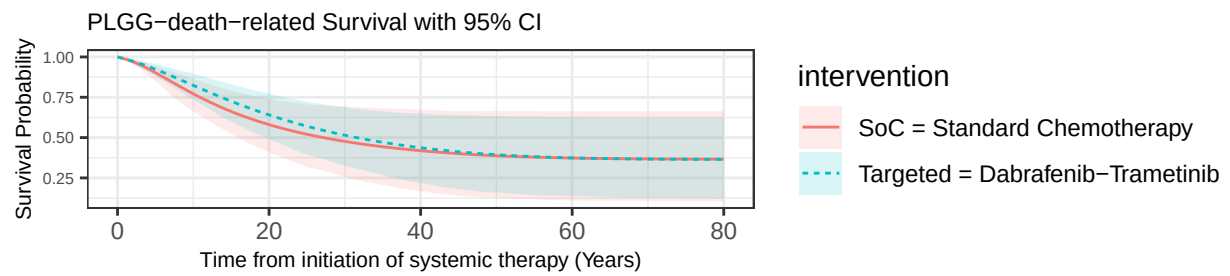
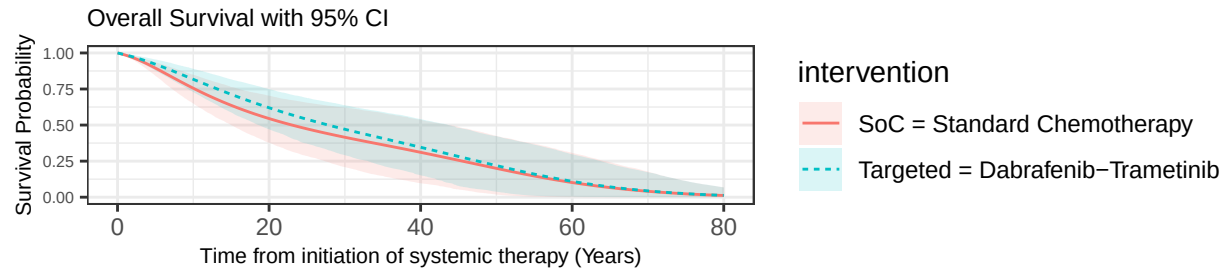


Trace Plot of PLGG States Across Cycles



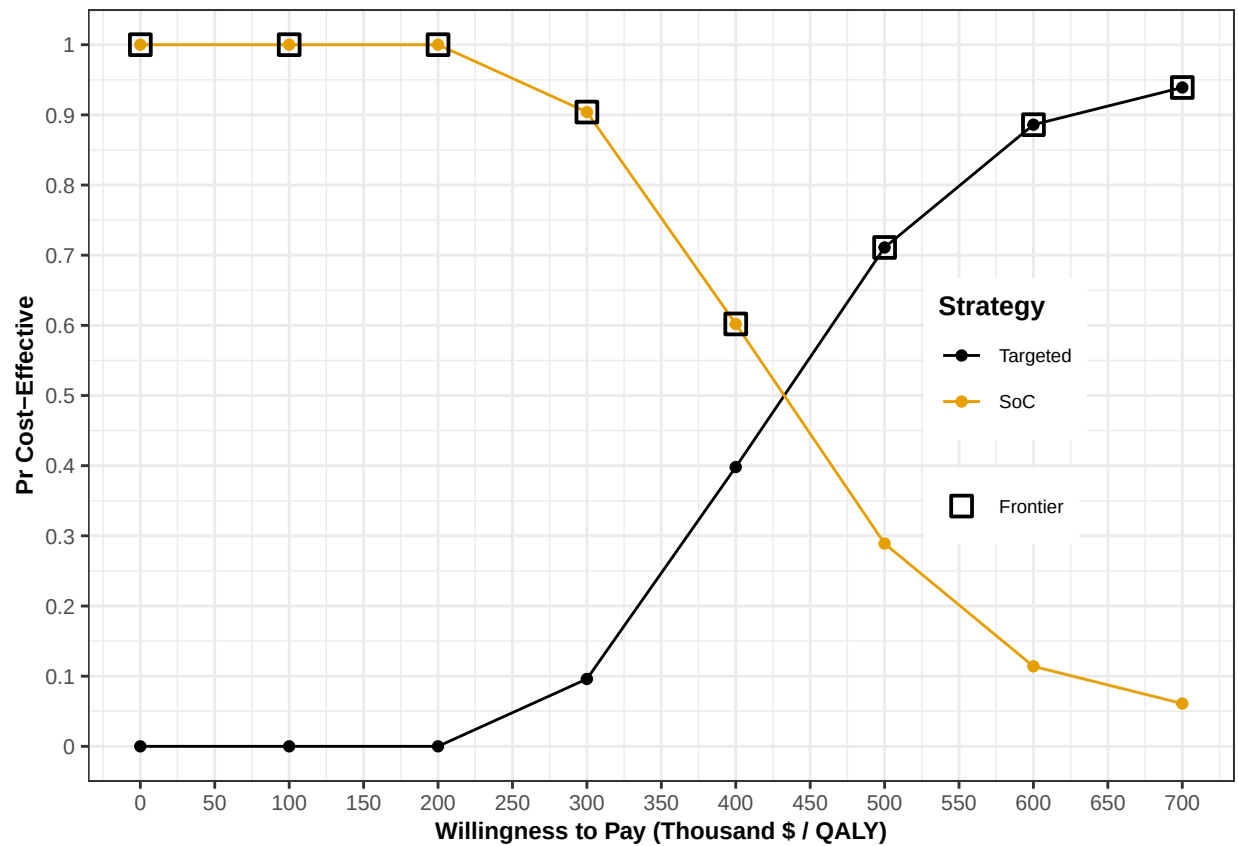
Cumulative Incidence of PLGG-related Late Effects Across Cycles



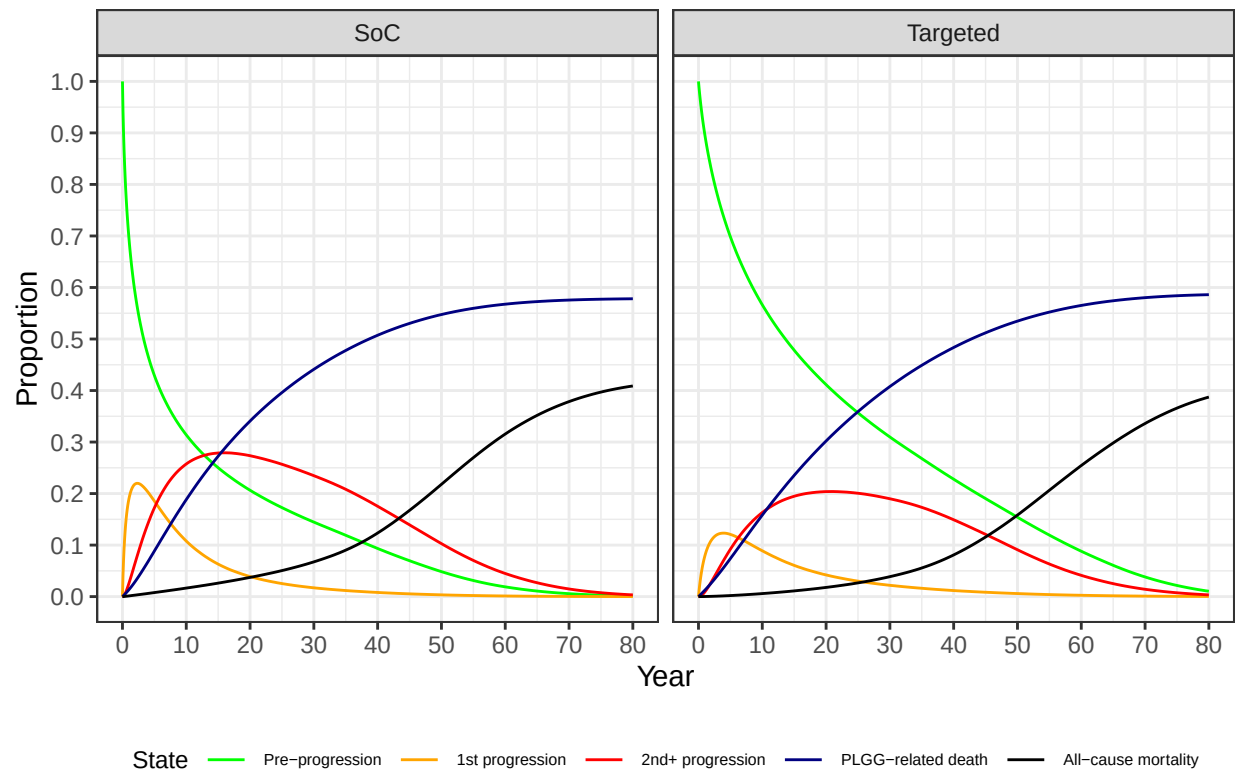


Investigator assessment, Targeted curve – Log-Normal, SoC curve – Log-Normal

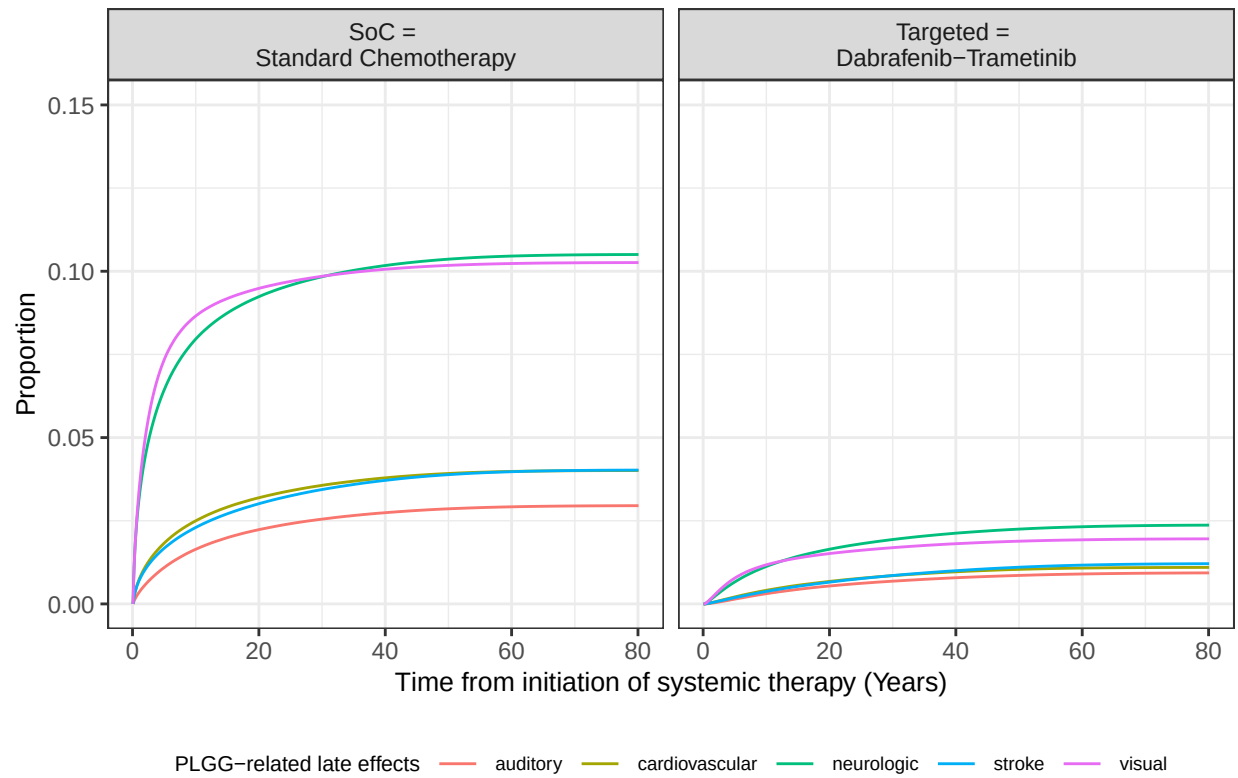
Variable	intervention	control	Delta
Life-years	35.79 (27.51;43.61)	31.67 (23.64;40.98)	4.12 (-0.92;11.62)
QALY	16.76 (13.05;20.47)	13.52 (10.18;17.13)	3.24 (0.85;6.63)
Total Cost	\$1,728,126 (\$1,195,492;\$2,373,887)	\$399,090 (\$375,734;\$429,754)	\$1,329,036 (\$803,928;\$1,977,259)
Non-Targeted PLGG related costs	\$172,134 (\$129,314;\$219,466)	\$238,222 (\$184,748;\$299,276)	\$-66,088 (\$-119,569;\$-21,788)
Late effect costs	\$77,267 (\$40,718;\$127,374)	\$124,150 (\$70,435;\$189,145)	\$-46,883 (\$-78,601;\$-15,893)
General population costs	\$65,756 (\$42,281;\$92,588)	\$36,718 (\$20,754;\$60,980)	\$29,038 (\$14,534;\$50,612)
Targeted costs	\$1,412,969 (\$879,150;\$2,074,561)	\$0 (\$0;\$0)	\$1,412,969 (\$879,150;\$2,074,561)
Radiation costs	\$0 (\$0;\$0)	\$0 (\$0;\$0)	\$0 (\$0;\$0)

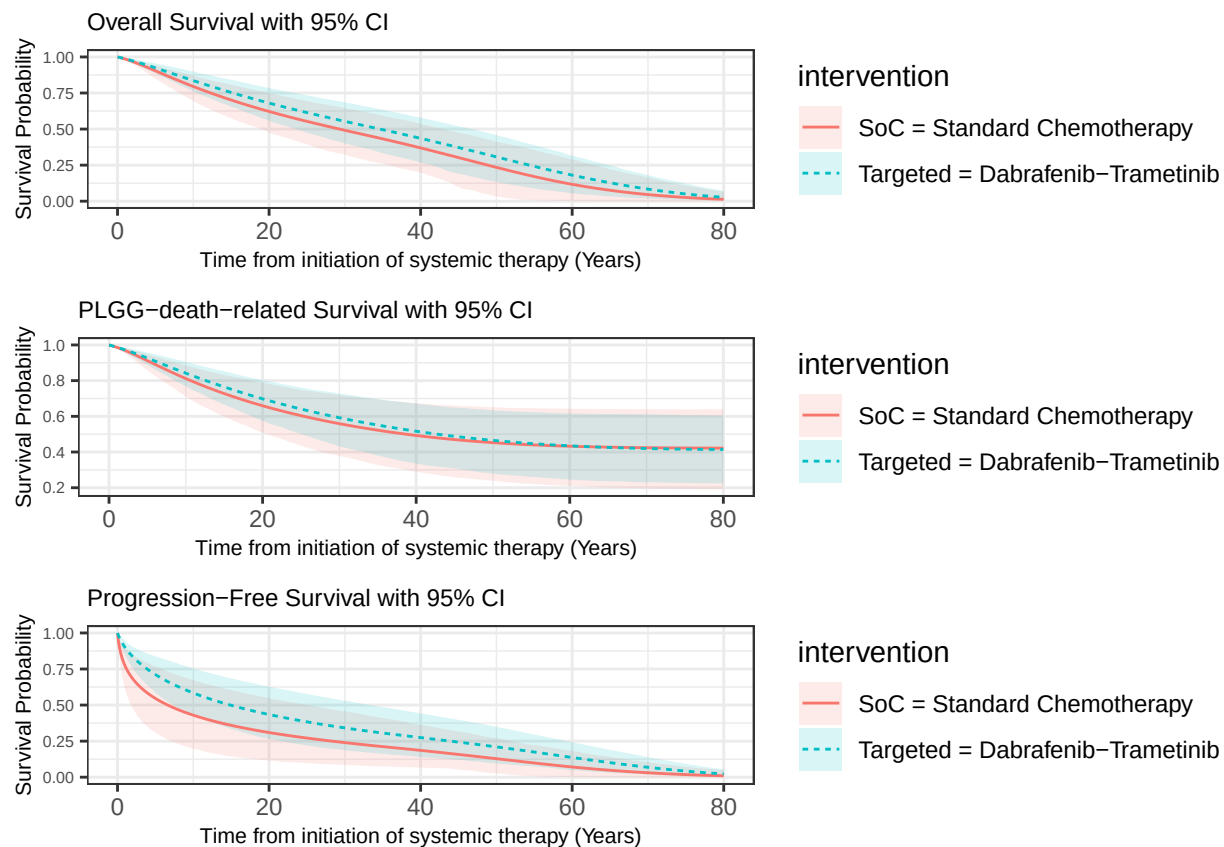


Trace Plot of PLGG States Across Cycles



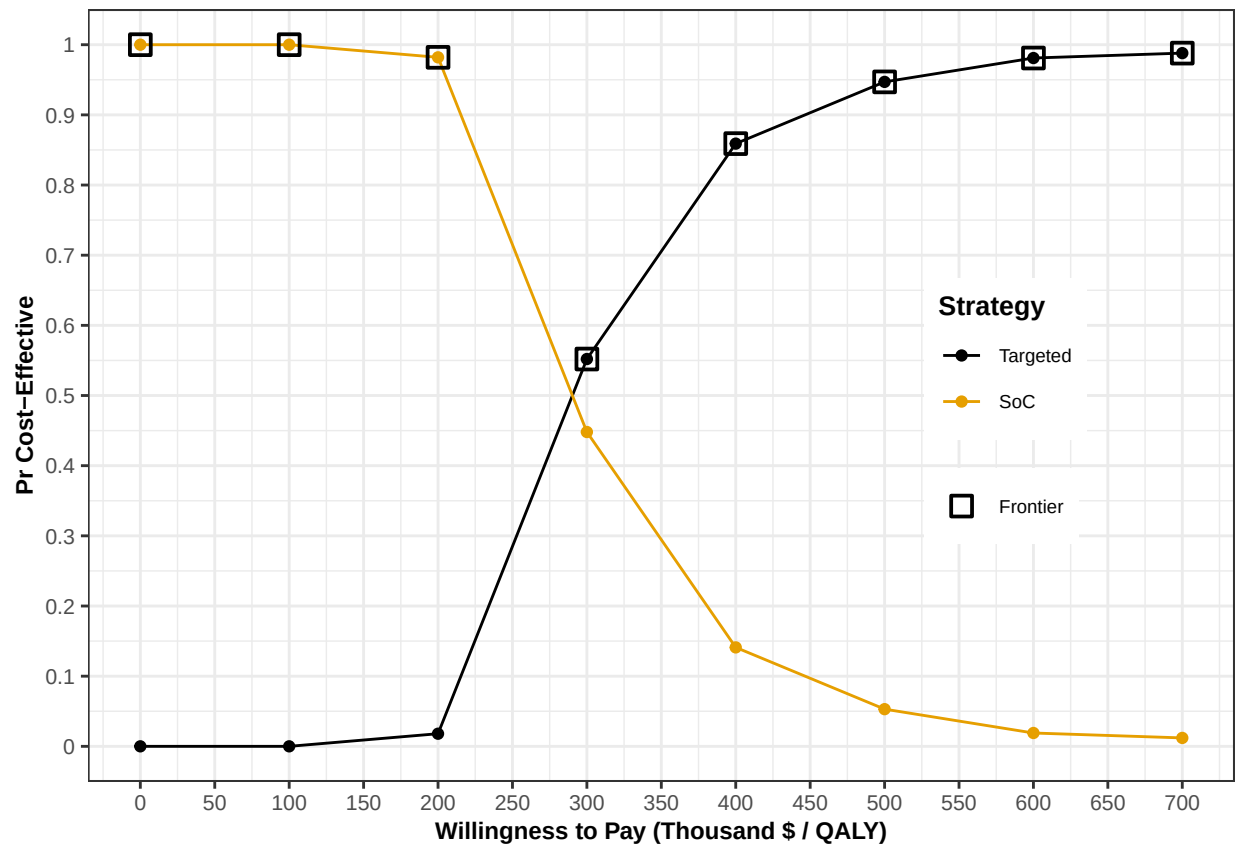
Cumulative Incidence of PLGG-related Late Effects Across Cycles



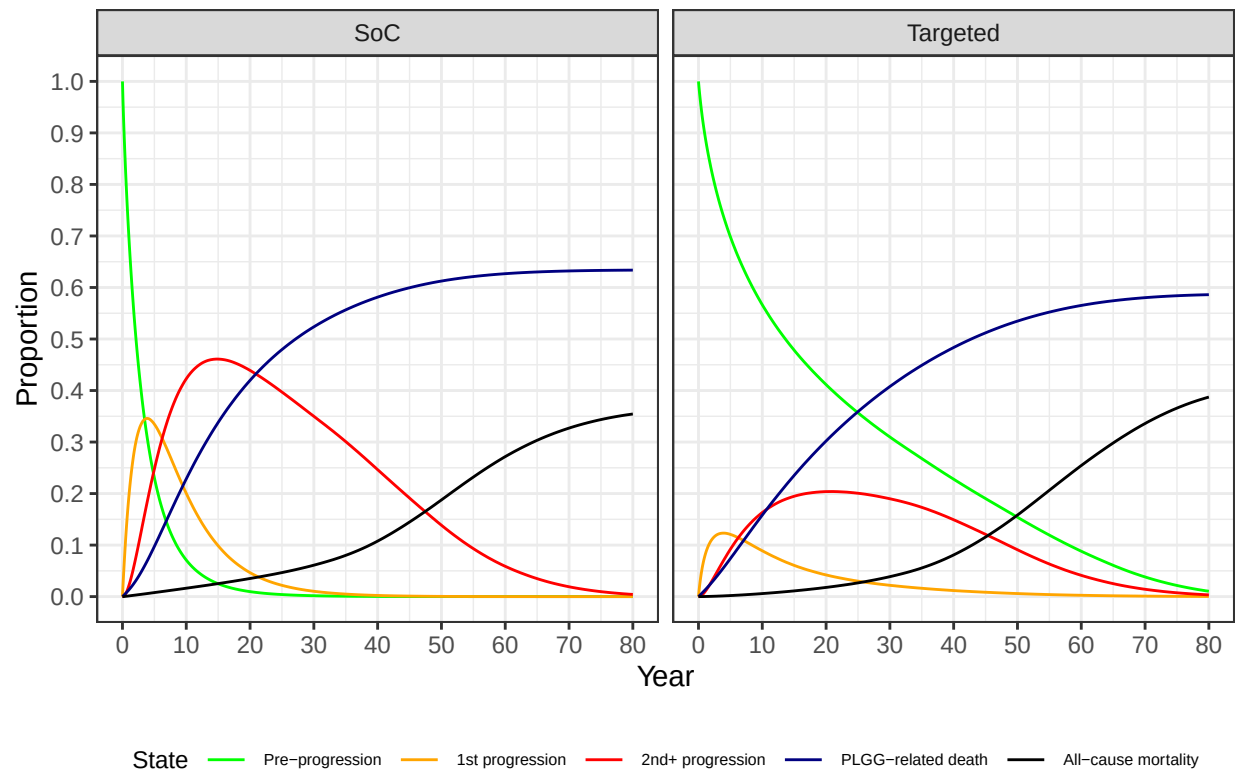


Investigator assessment, Targeted curve – Log-Normal, SoC curve – Exponential

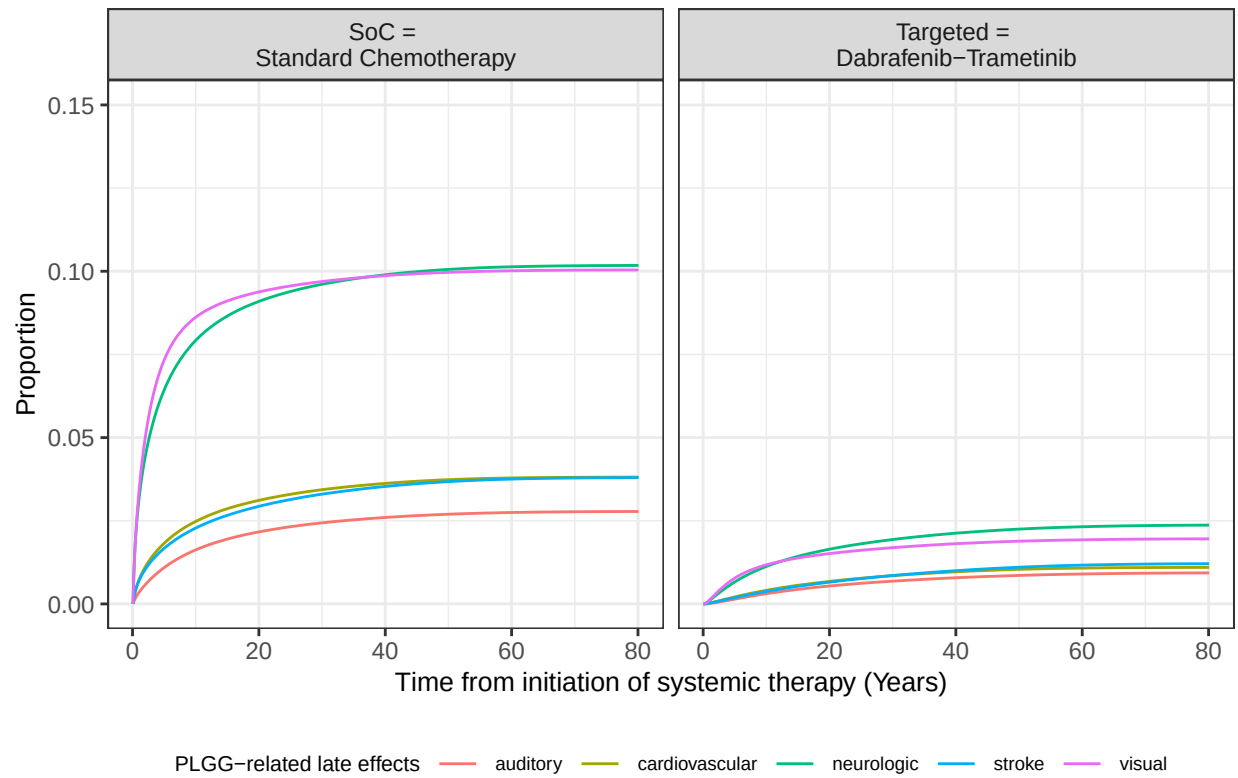
Variable	intervention	control	Delta
Life-years	35.79 (27.51;43.61)	28.62 (19.25;39.7)	7.17 (1.17;15.7)
QALY	16.76 (13.05;20.47)	12.28 (8.51;16.83)	4.48 (1.83;8.36)
Total Cost	\$1,728,126 (\$1,195,492;\$2,373,887)	\$415,564 (\$386,410;\$449,808)	\$1,312,562 (\$787,687;\$1,961,090)
Non-Targeted PLGG related costs	\$172,134 (\$129,314;\$219,466)	\$273,899 (\$210,008;\$334,620)	\$-101,765 (\$-160,633;\$-59,664)
Late effect costs	\$77,267 (\$40,718;\$127,374)	\$109,507 (\$47,829;\$189,429)	\$-32,240 (\$-73,617;\$5,608)
General population costs	\$65,756 (\$42,281;\$92,588)	\$32,158 (\$15,260;\$60,585)	\$33,597 (\$17,738;\$57,135)
Targeted costs	\$1,412,969 (\$879,150;\$2,074,561)	\$0 (\$0;\$0)	\$1,412,969 (\$879,150;\$2,074,561)
Radiation costs	\$0 (\$0;\$0)	\$0 (\$0;\$0)	\$0 (\$0;\$0)

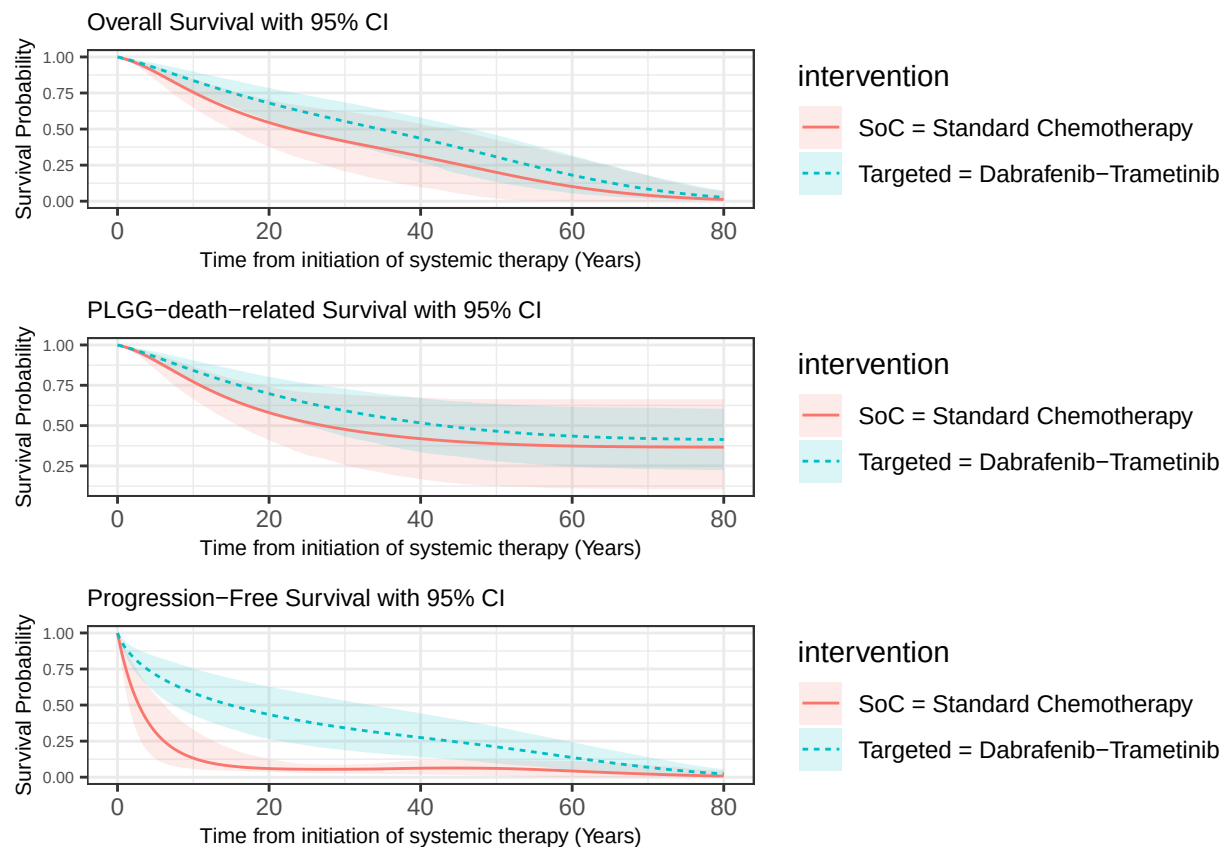


Trace Plot of PLGG States Across Cycles



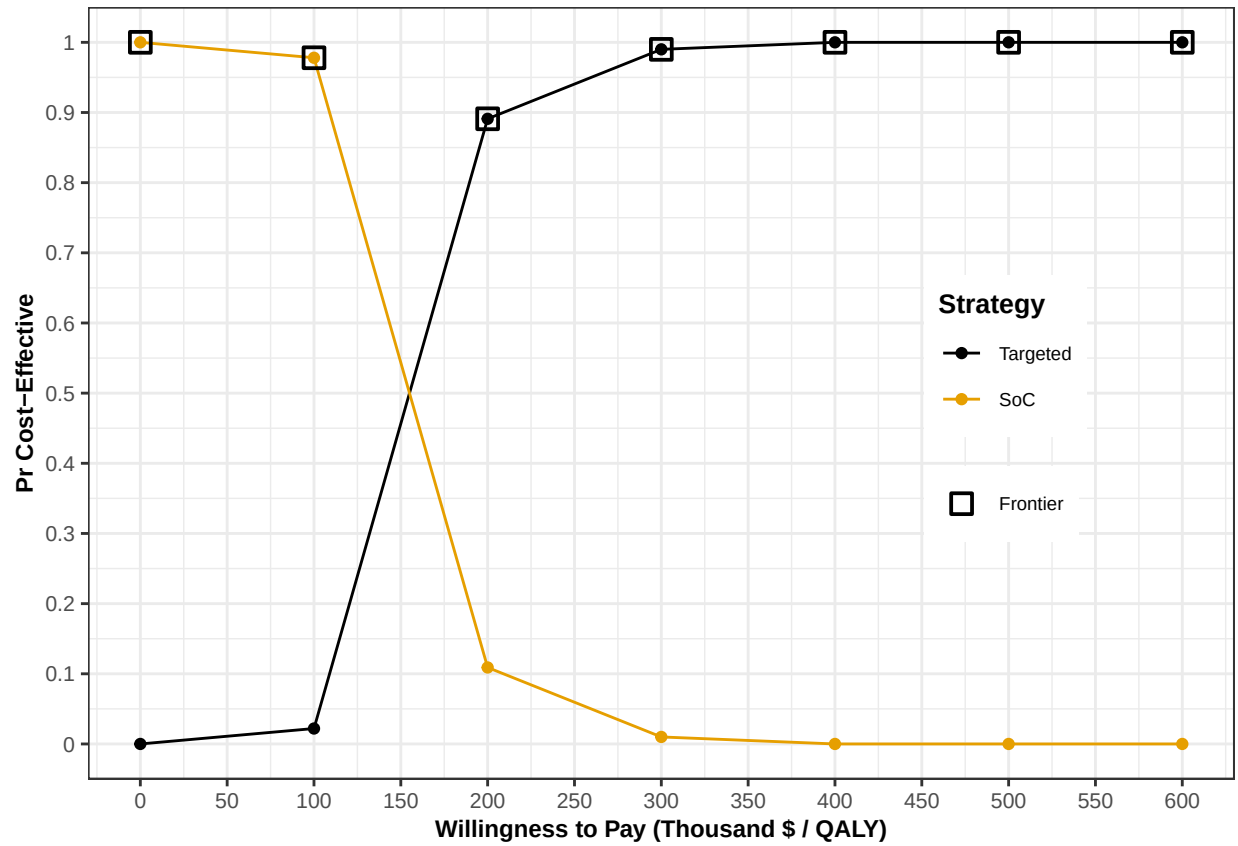
Cumulative Incidence of PLGG-related Late Effects Across Cycles



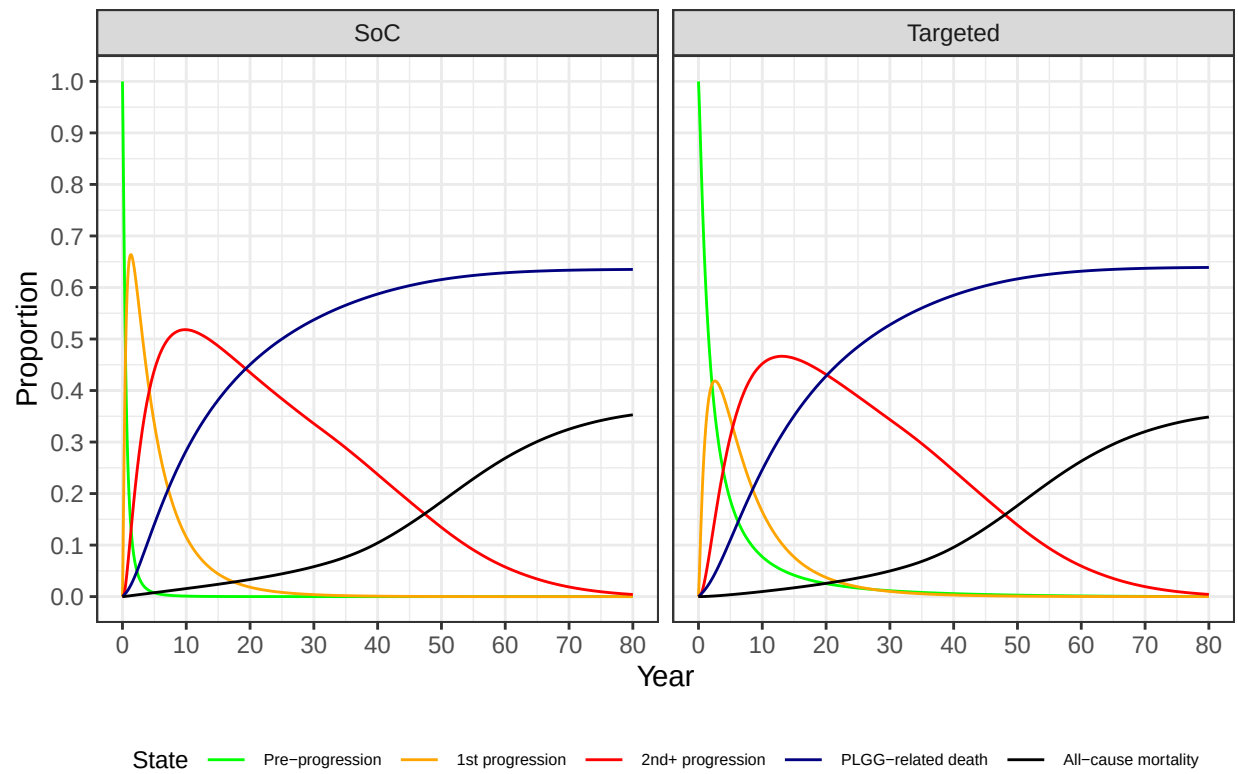


Independent reviewer, Targeted curve – Exponential, SoC curve – exponential

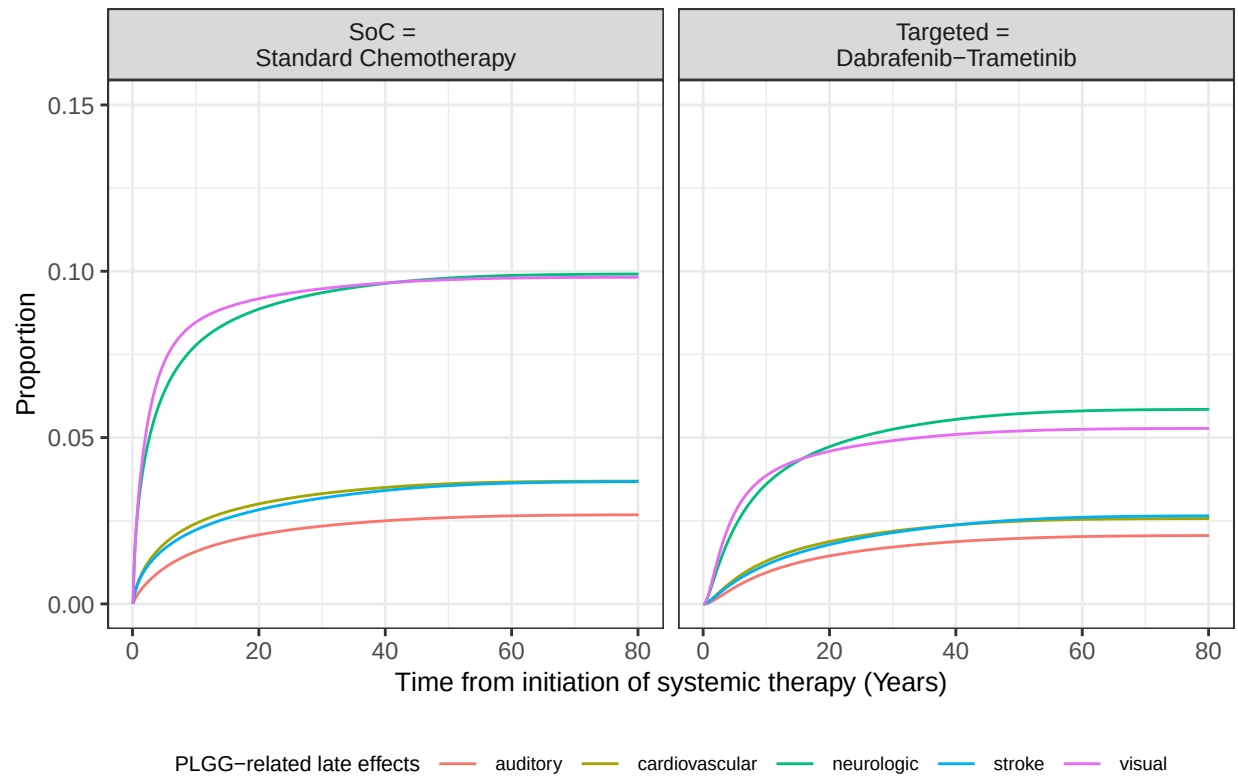
Variable	intervention	control	Delta
Life-years	28.8 (19.36;40.17)	27.62 (17.45;39.75)	1.18 (-0.02;2.81)
QALY	12.99 (8.83;17.85)	11.84 (7.56;16.96)	1.15 (0.61;1.96)
Total Cost	\$599,383 (\$507,984;\$762,956)	\$421,510 (\$389,810;\$459,257)	\$177,873 (\$81,564;\$335,724)
Non-Targeted PLGG related costs	\$241,279 (\$178,440;\$303,448)	\$281,994 (\$214,592;\$344,052)	\$-40,715 (\$-57,476;\$-29,713)
Late effect costs	\$90,244 (\$30,548;\$169,169)	\$108,161 (\$41,629;\$194,270)	\$-17,917 (\$-31,183;\$-8,710)
General population costs	\$39,447 (\$19,765;\$71,143)	\$31,355 (\$13,376;\$61,817)	\$8,092 (\$4,922;\$13,057)
Targeted costs	\$228,413 (\$119,694;\$390,346)	\$0 (\$0;\$0)	\$228,413 (\$119,694;\$390,346)
Radiation costs	\$0 (\$0;\$0)	\$0 (\$0;\$0)	\$0 (\$0;\$0)

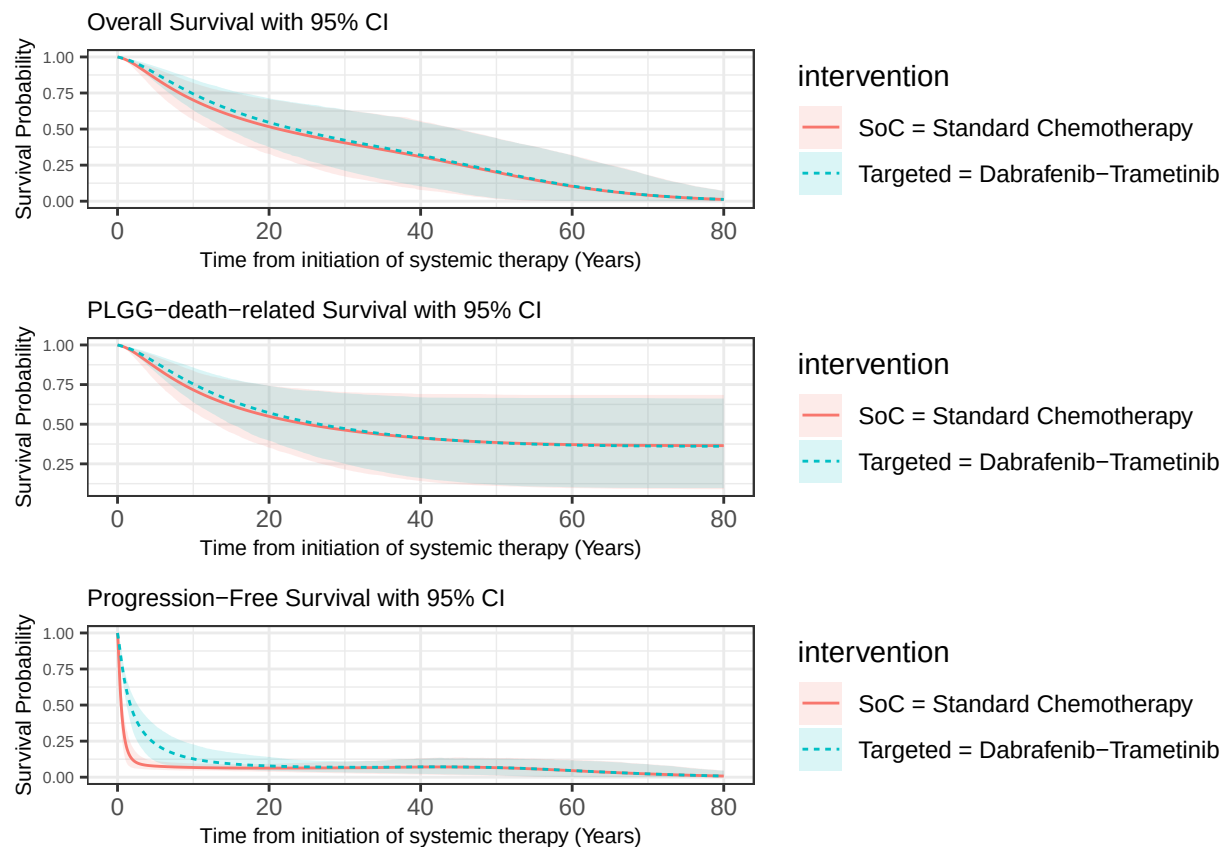


Trace Plot of PLGG States Across Cycles



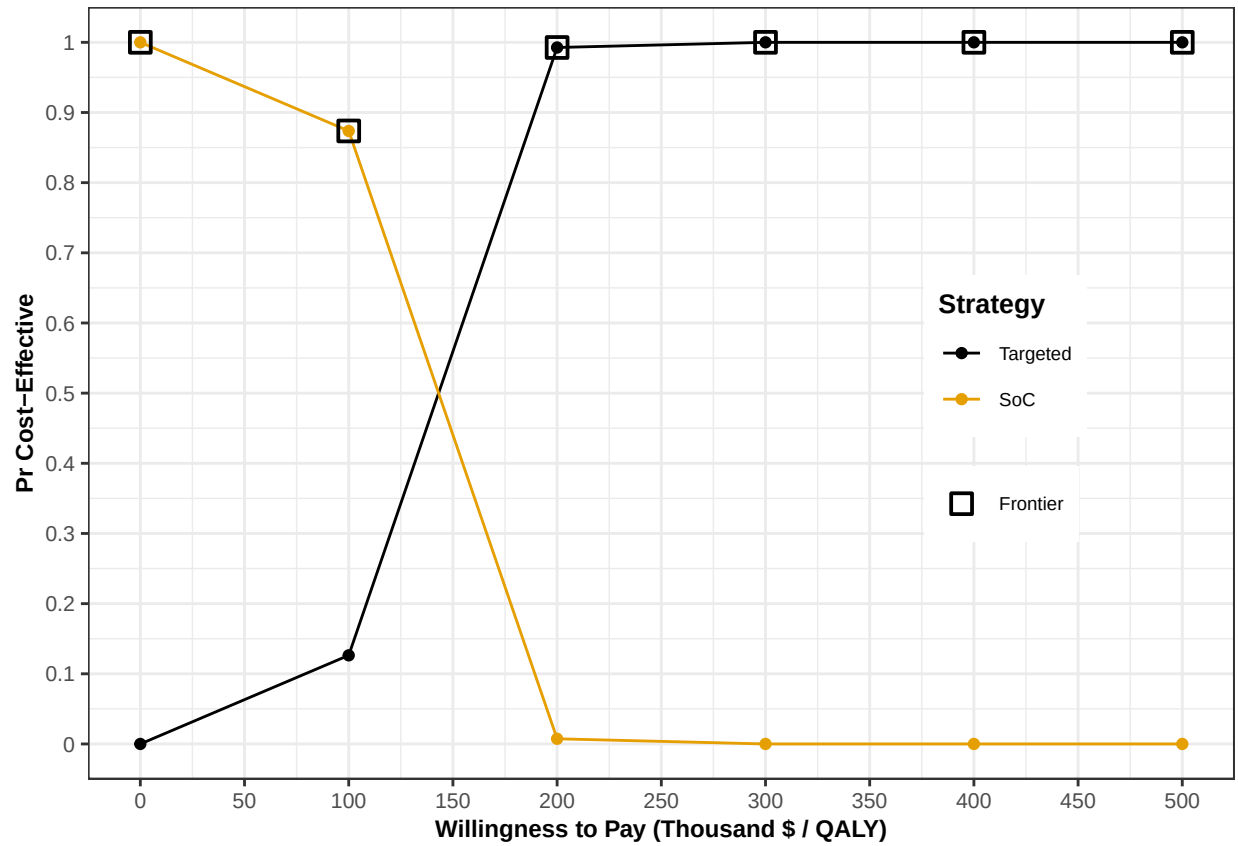
Cumulative Incidence of PLGG-related Late Effects Across Cycles



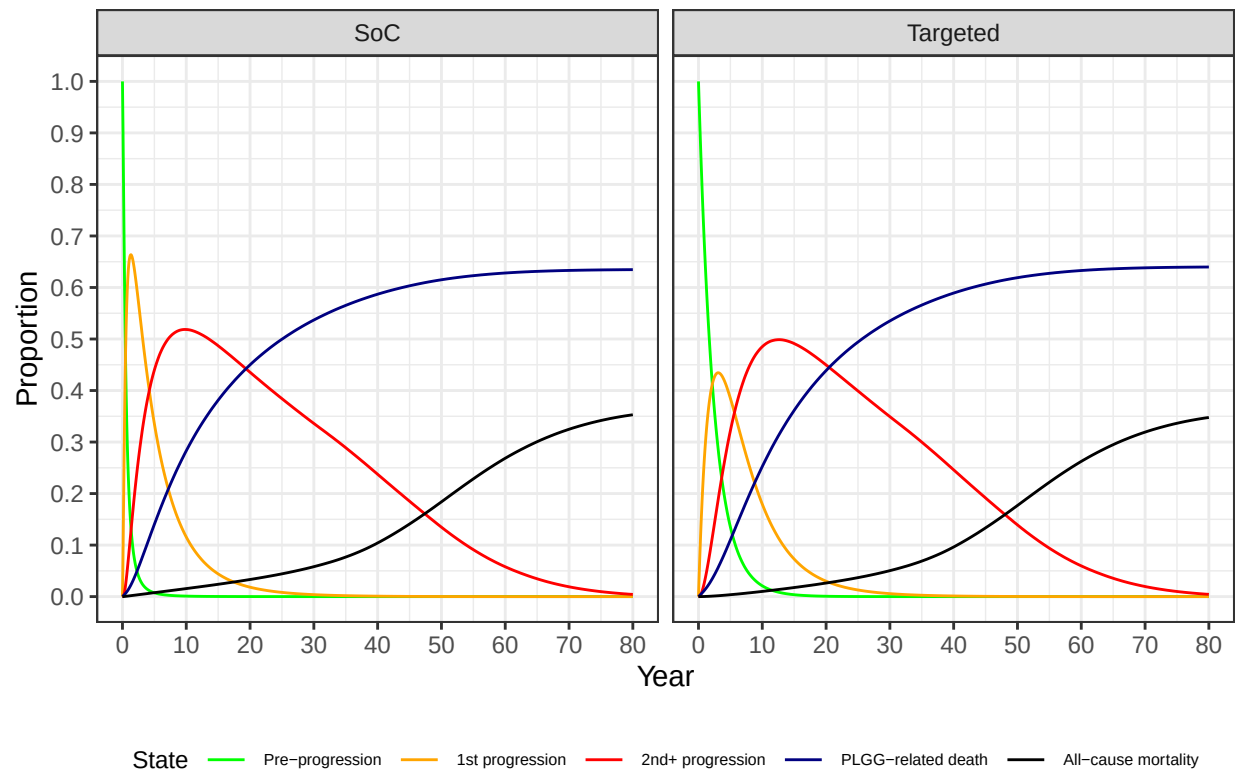


Independent reviewer, Targeted curve – Log-Normal, SoC curve – Log-Normal

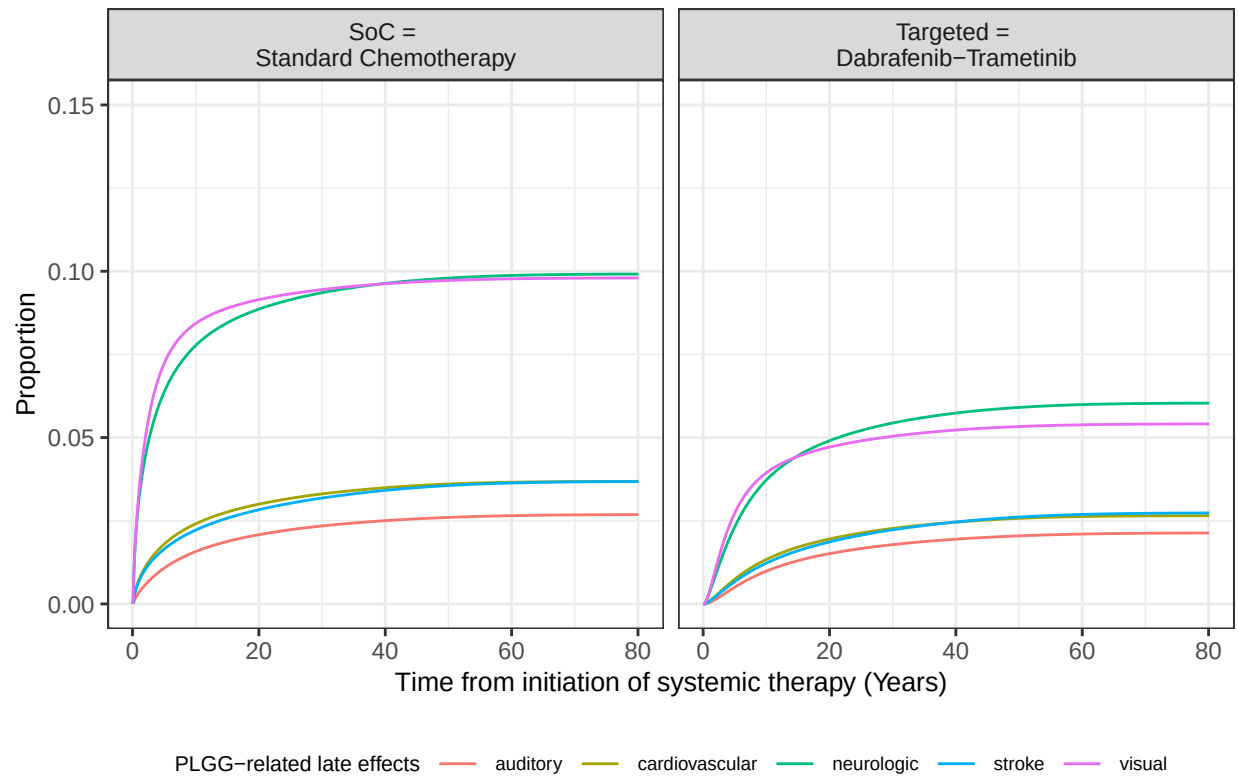
Variable	intervention	control	Delta
Life-years	28.46 (18.57;40.19)	27.65 (17.45;40.01)	0.81 (-0.02;1.79)
QALY	12.81 (8.5;17.83)	11.86 (7.56;17.02)	0.95 (0.57;1.49)
Total Cost	\$536,542 (\$487,396;\$610,206)	\$421,567 (\$389,726;\$459,810)	\$114,976 (\$74,951;\$188,944)
Non-Targeted PLGG related costs	\$246,083 (\$177,084;\$308,393)	\$281,970 (\$214,432;\$344,176)	\$-35,887 (\$-47,461;\$-28,007)
Late effect costs	\$90,904 (\$29,475;\$173,037)	\$108,133 (\$41,550;\$194,301)	\$-17,229 (\$-29,091;\$-9,409)
General population costs	\$38,314 (\$18,438;\$71,855)	\$31,464 (\$13,358;\$62,417)	\$6,850 (\$4,299;\$10,791)
Targeted costs	\$161,241 (\$115,498;\$247,741)	\$0 (\$0;\$0)	\$161,241 (\$115,498;\$247,741)
Radiation costs	\$0 (\$0;\$0)	\$0 (\$0;\$0)	\$0 (\$0;\$0)

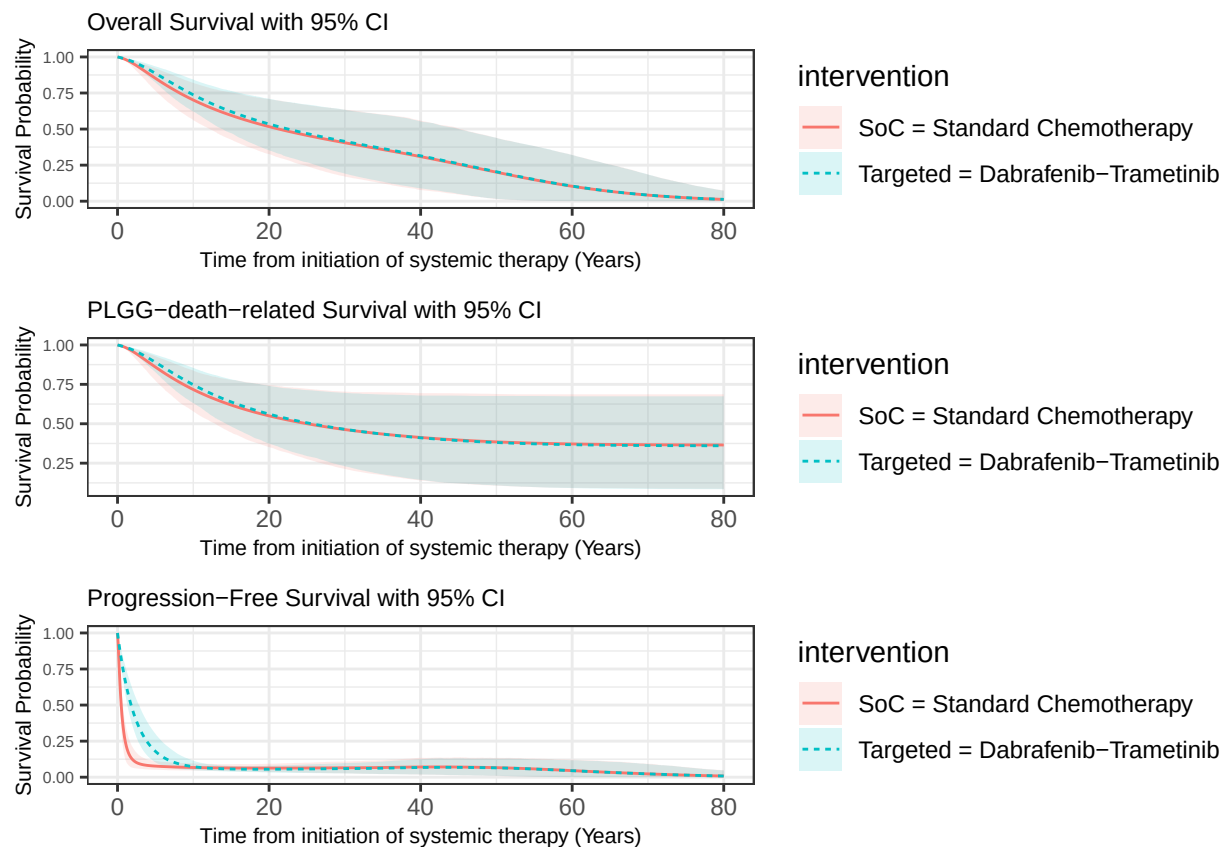


Trace Plot of PLGG States Across Cycles



Cumulative Incidence of PLGG-related Late Effects Across Cycles



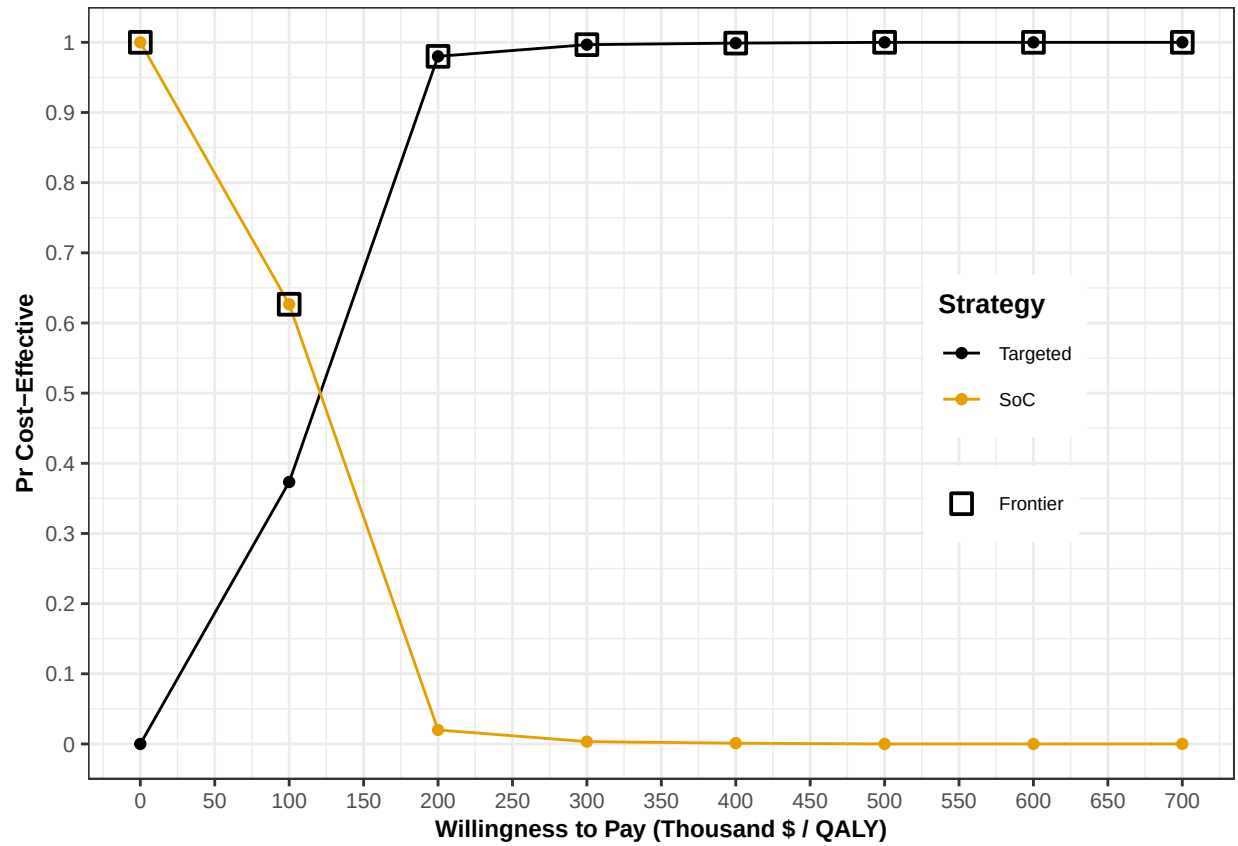


Scenario #1c

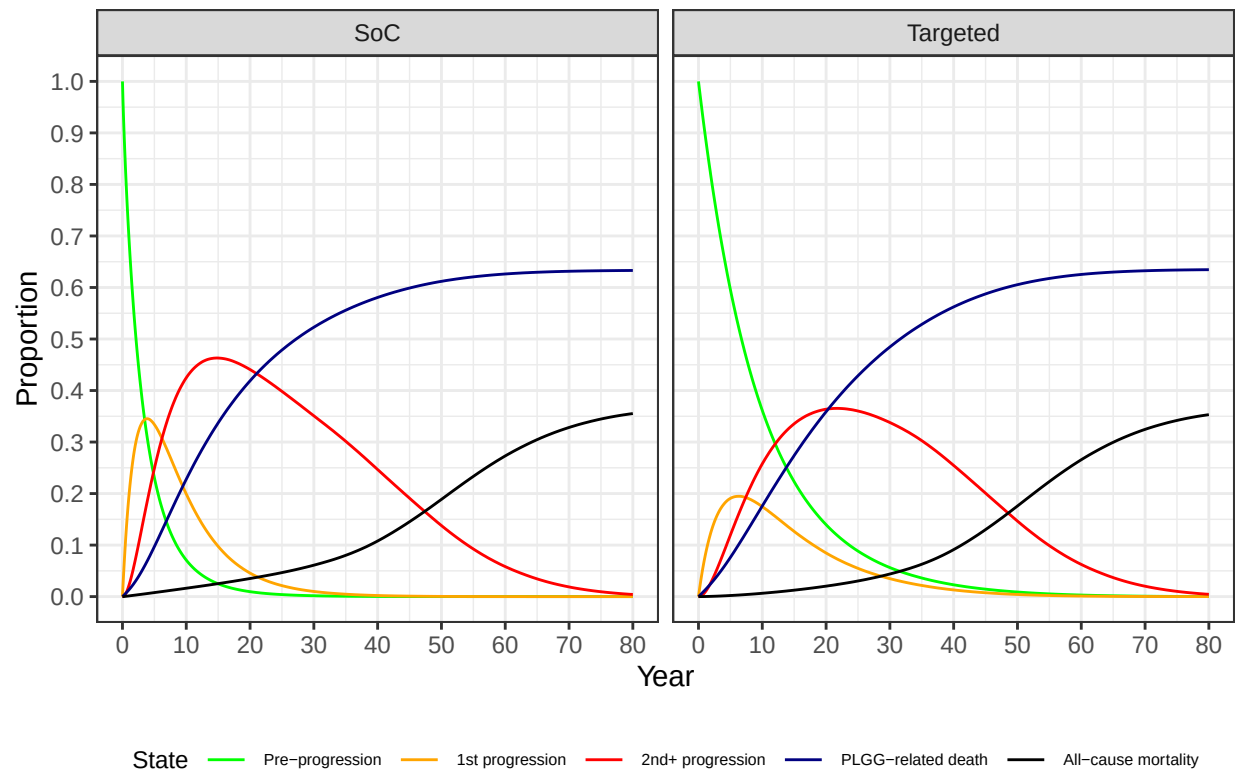
- Combo vs. chemo as first-line treatment
- Lifetime duration of combo therapy
- 0% radiation at progression (Post-2014 SickKids data indicates that no patients who received chemo and progressed were radiated)
- **50% combo price reduction**
- Discount rate: 1.5
- Data source for controls & exposed: Investigator assessment - Bouffet et al., NEJM 2023.
- **Post-progression risk for subsequent progression = pre-progression risk for first progression in combo arm
- 10,000 individuals
- 50 simulation runs

Investigator assessment, Targeted curve – exponential, SoC curve – exponential

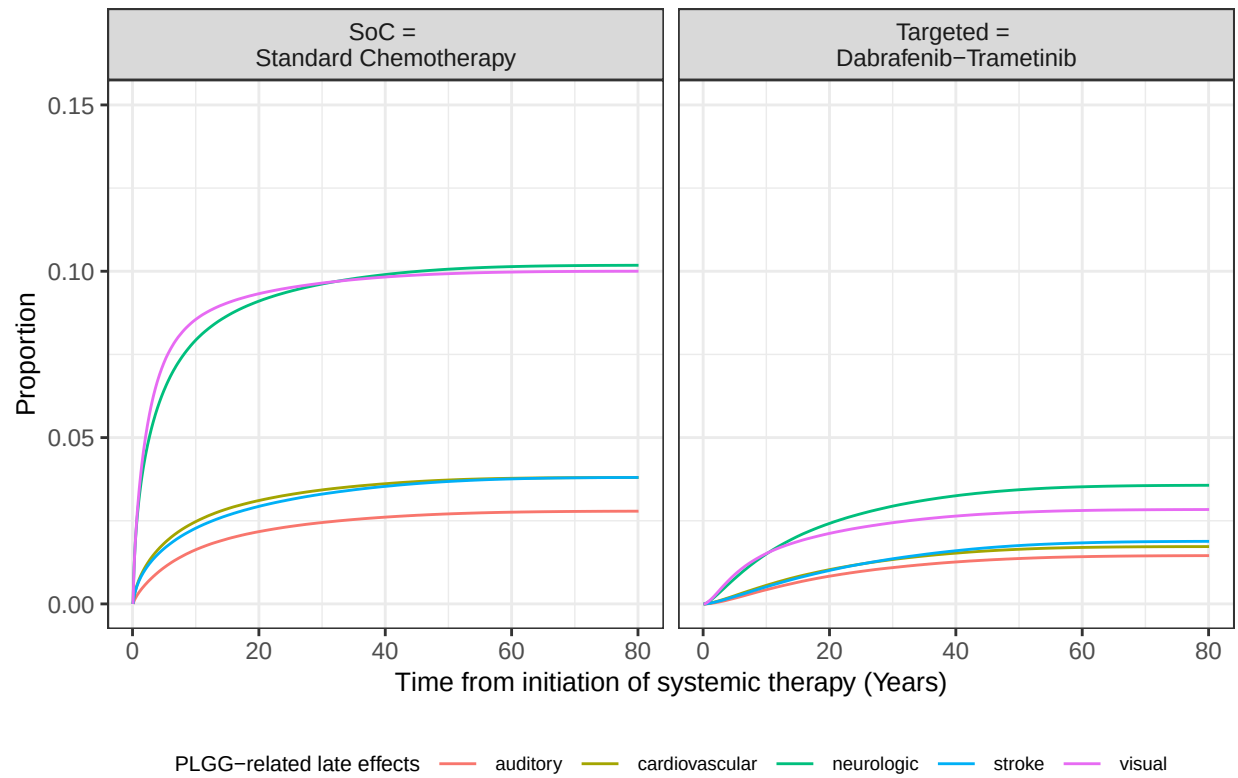
Variable	intervention	control	Delta
Life-years	31.21 (22.6;40.73)	28.63 (19.53;39.76)	2.59 (0.09;7.36)
QALY	14.49 (10.58;18.73)	12.29 (8.51;16.93)	2.2 (0.94;4.31)
Total Cost	\$650,442 (\$533,733;\$837,625)	\$415,874 (\$387,901;\$449,768)	\$234,567 (\$122,863;\$421,768)
Non-Targeted PLGG related costs	\$199,635 (\$140,988;\$259,563)	\$273,913 (\$209,766;\$334,189)	\$-74,277 (\$-107,606;\$-47,767)
Late effect costs	\$81,208 (\$29,301;\$146,895)	\$109,844 (\$47,983;\$189,664)	\$-28,636 (\$-51,187;\$-9,153)
General population costs	\$48,836 (\$28,482;\$81,111)	\$32,118 (\$15,329;\$61,113)	\$16,718 (\$9,770;\$27,498)
Targeted costs	\$320,763 (\$193,362;\$518,108)	\$0 (\$0;\$0)	\$320,763 (\$193,362;\$518,108)
Radiation costs	\$0 (\$0;\$0)	\$0 (\$0;\$0)	\$0 (\$0;\$0)

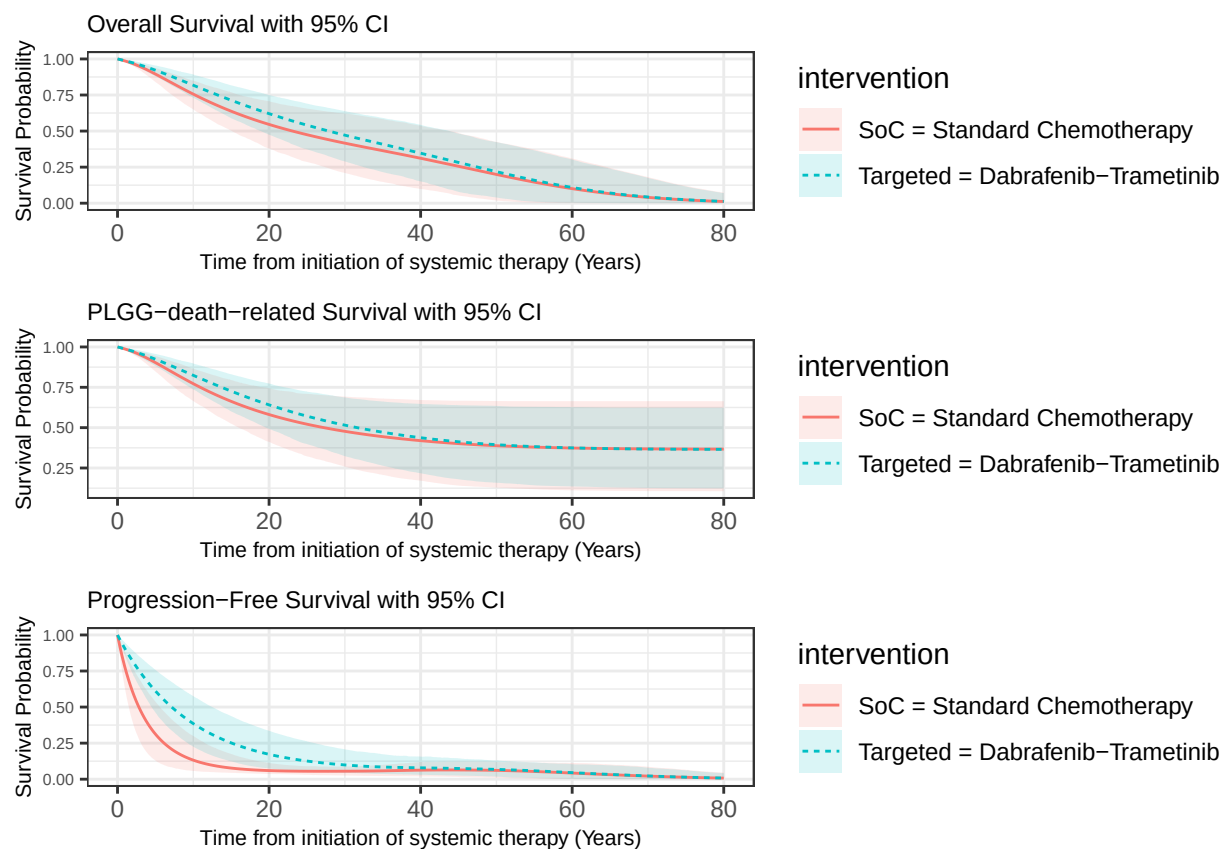


Trace Plot of PLGG States Across Cycles



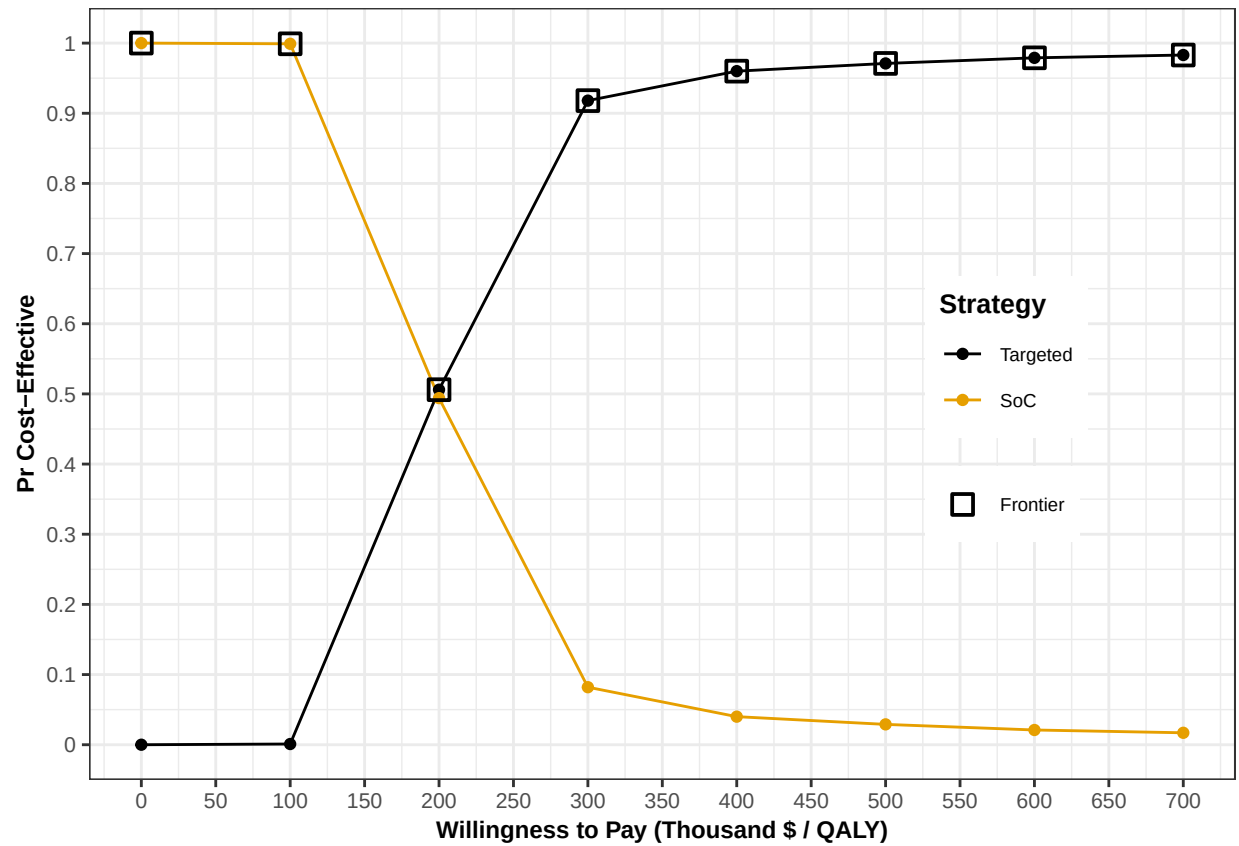
Cumulative Incidence of PLGG-related Late Effects Across Cycles



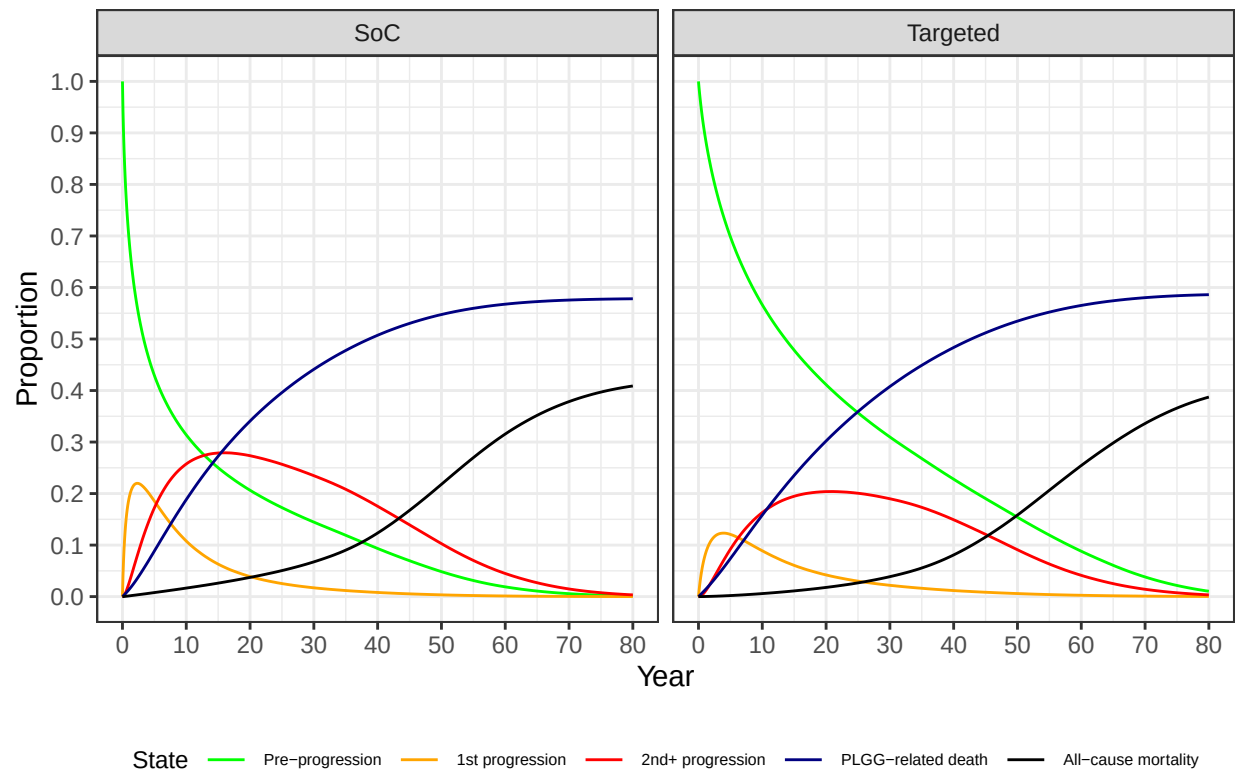


Investigator assessment, Targeted curve – lognormal, SoC curve – lognormal

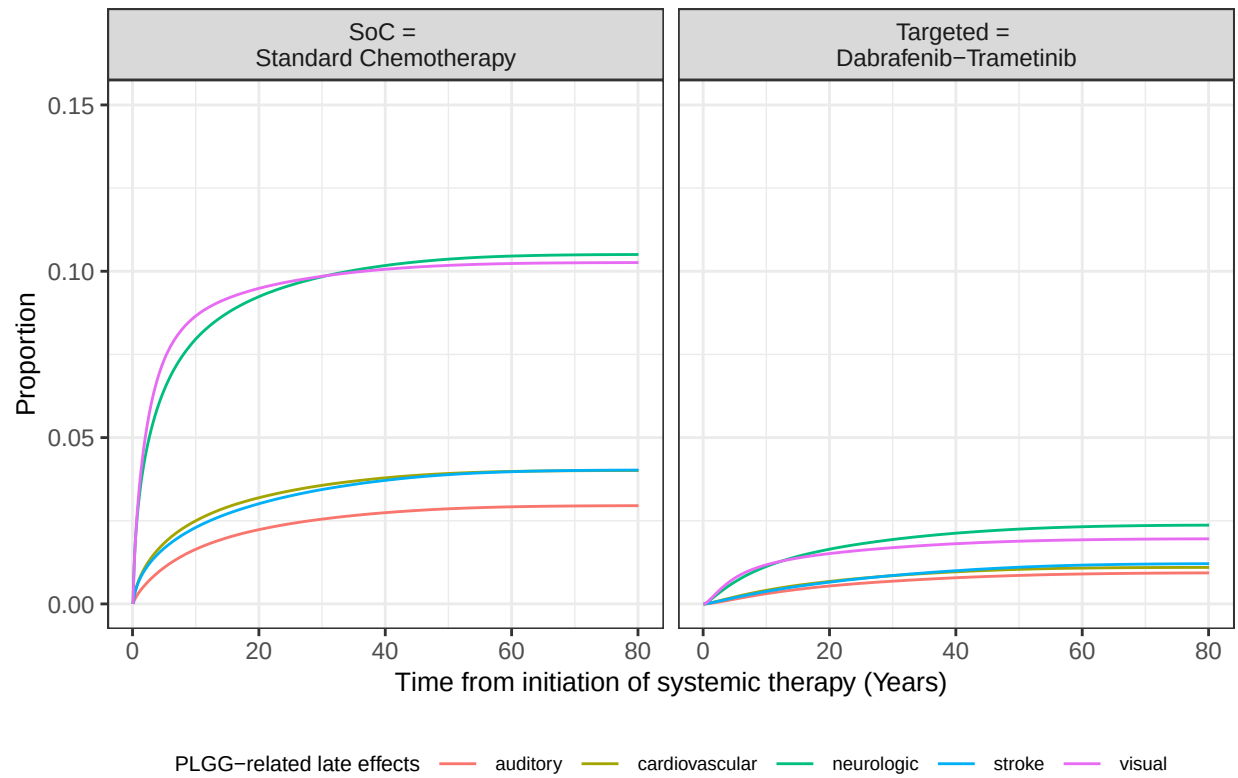
Variable	intervention	control	Delta
Life-years	35.79 (27.51;43.61)	31.67 (23.64;40.98)	4.12 (-0.92;11.62)
QALY	16.76 (13.05;20.47)	13.52 (10.18;17.13)	3.24 (0.85;6.63)
Total Cost	\$1,021,641 (\$757,913;\$1,339,888)	\$399,090 (\$375,734;\$429,754)	\$622,552 (\$365,306;\$944,835)
Non-Targeted PLGG related costs	\$172,134 (\$129,314;\$219,466)	\$238,222 (\$184,748;\$299,276)	\$-66,088 (\$-119,569;\$-21,788)
Late effect costs	\$77,267 (\$40,718;\$127,374)	\$124,150 (\$70,435;\$189,145)	\$-46,883 (\$-78,601;\$-15,893)
General population costs	\$65,756 (\$42,281;\$92,588)	\$36,718 (\$20,754;\$60,980)	\$29,038 (\$14,534;\$50,612)
Targeted costs	\$706,484 (\$439,575;\$1,037,280)	\$0 (\$0;\$0)	\$706,484 (\$439,575;\$1,037,280)
Radiation costs	\$0 (\$0;\$0)	\$0 (\$0;\$0)	\$0 (\$0;\$0)

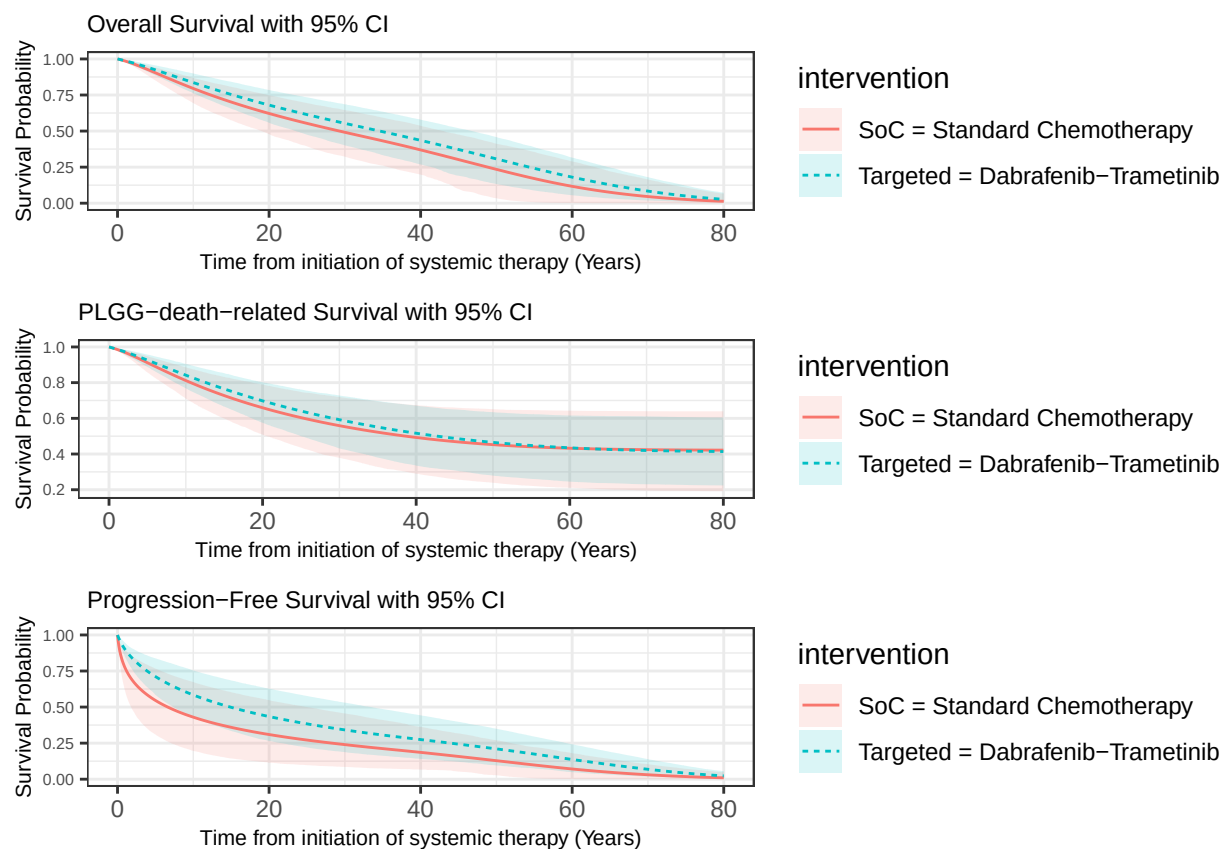


Trace Plot of PLGG States Across Cycles



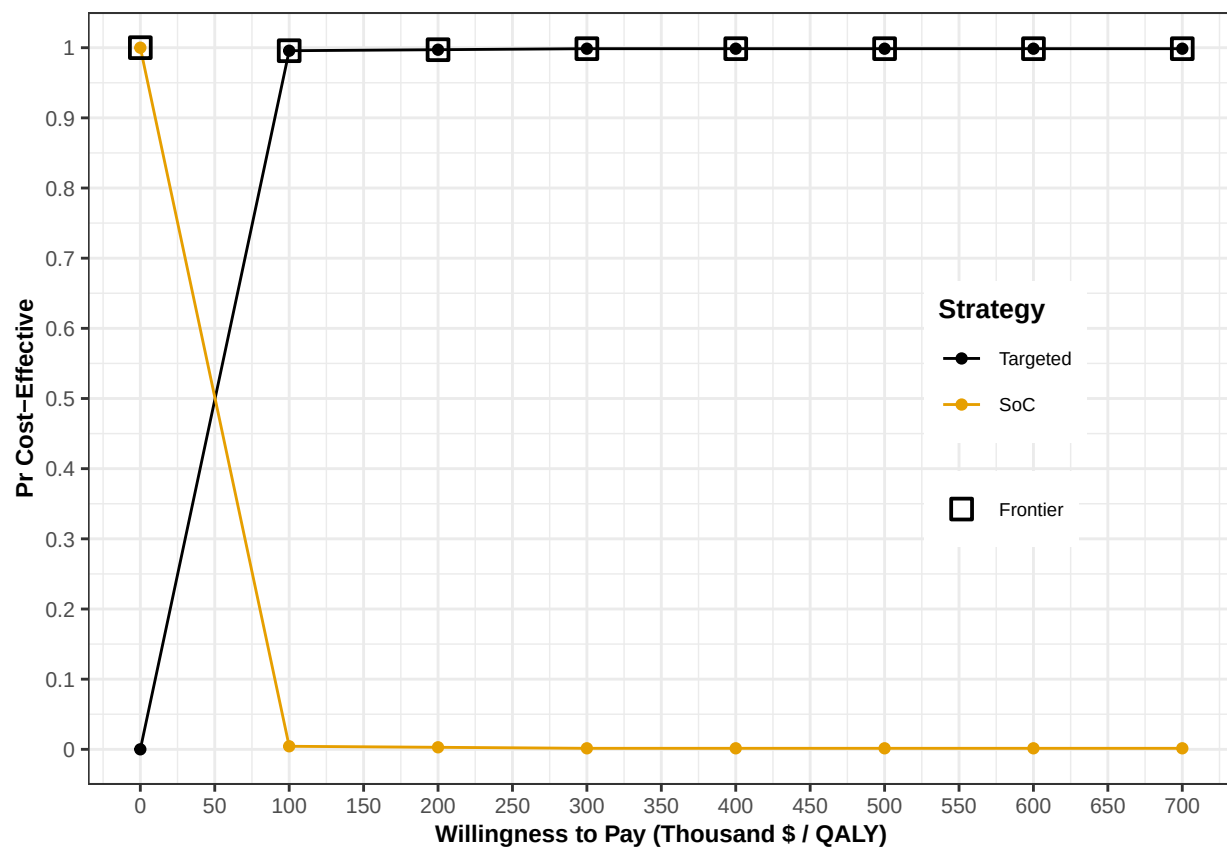
Cumulative Incidence of PLGG-related Late Effects Across Cycles



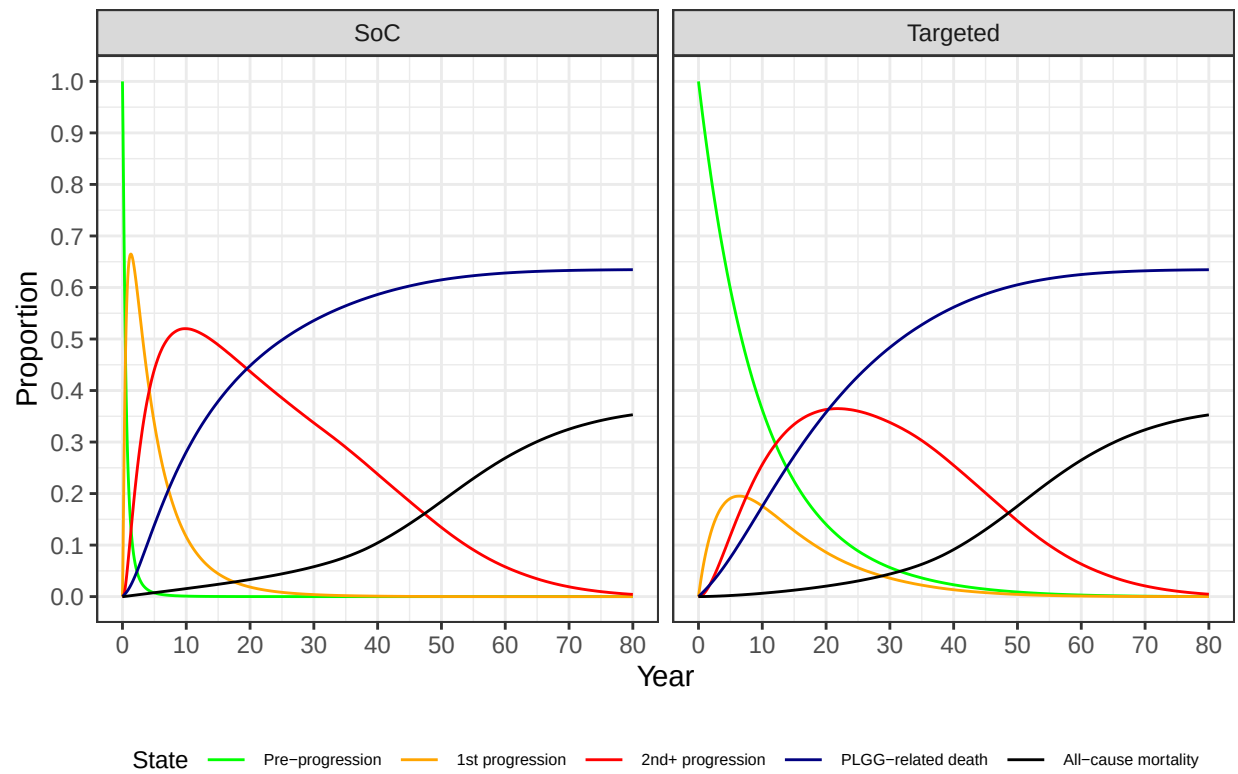


Investigator assessment, Targeted curve – exponential, SoC curve – exponential 75% discount

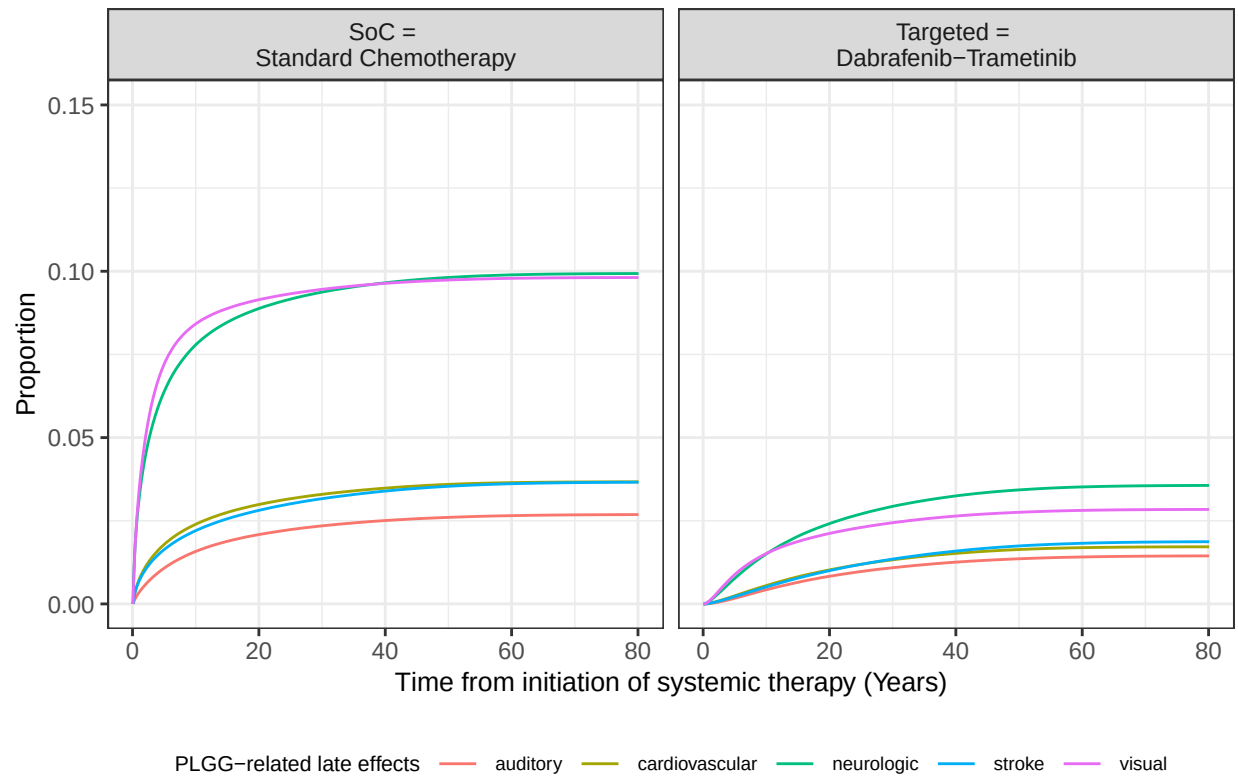
Variable	intervention	control	Delta
Life-years	31.27 (22.56;41.24)	27.7 (17.12;40.94)	3.57 (-0.5;9.02)
QALY	14.52 (10.53;18.86)	11.88 (7.53;17.22)	2.63 (1.03;4.94)
Total Cost	\$490,228 (\$433,250;\$577,045)	\$421,813 (\$389,141;\$459,638)	\$68,415 (\$18,166;\$159,055)
Non-Targeted PLGG related costs	\$199,485 (\$139,725;\$258,869)	\$282,073 (\$212,272;\$344,177)	-\$82,589 (\$-119,525;\$-59,904)
Late effect costs	\$80,929 (\$29,011;\$148,630)	\$108,230 (\$41,415;\$196,757)	-\$27,301 (\$-53,450;\$-3,748)
General population costs	\$49,064 (\$28,237;\$81,520)	\$31,509 (\$12,961;\$63,136)	\$17,554 (\$9,969;\$29,773)
Targeted costs	\$160,751 (\$101,011;\$257,819)	\$0 (\$0;\$0)	\$160,751 (\$101,011;\$257,819)
Radiation costs	\$0 (\$0;\$0)	\$0 (\$0;\$0)	\$0 (\$0;\$0)

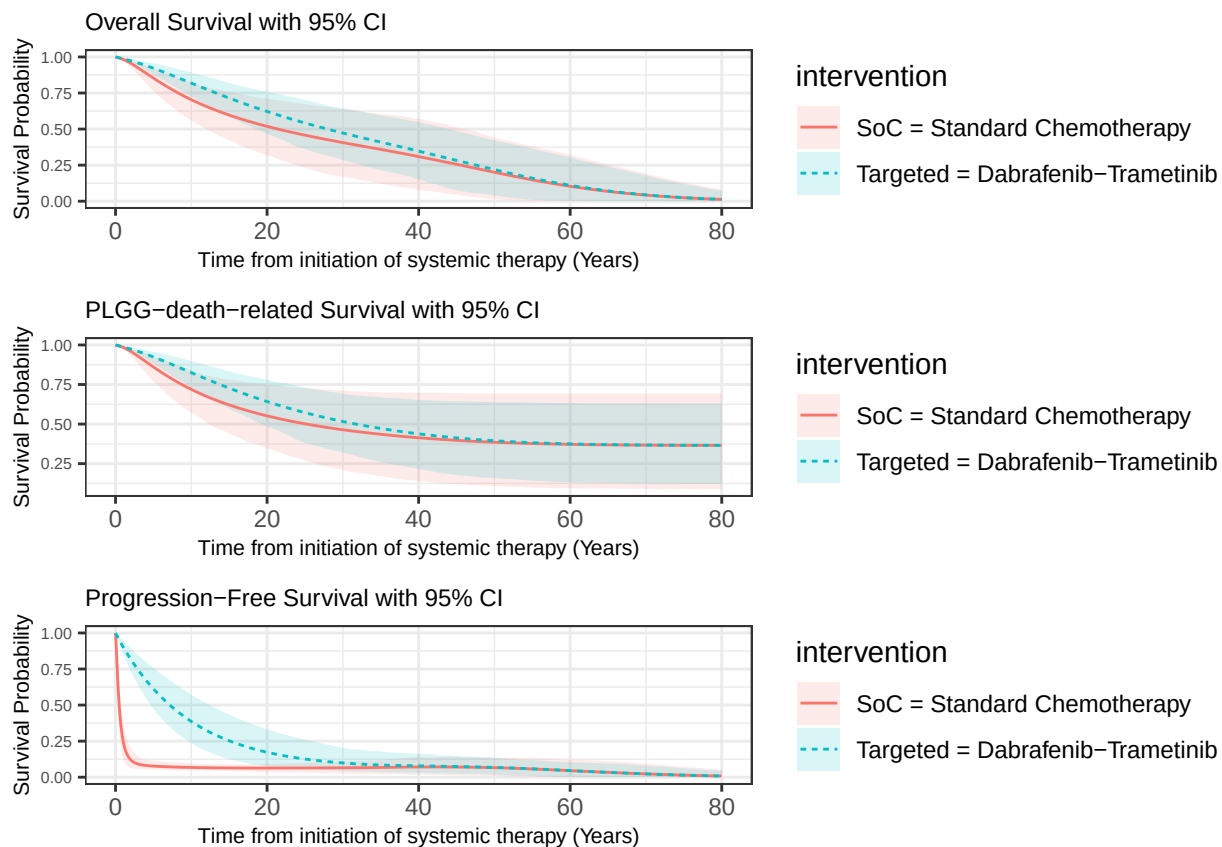


Trace Plot of PLGG States Across Cycles



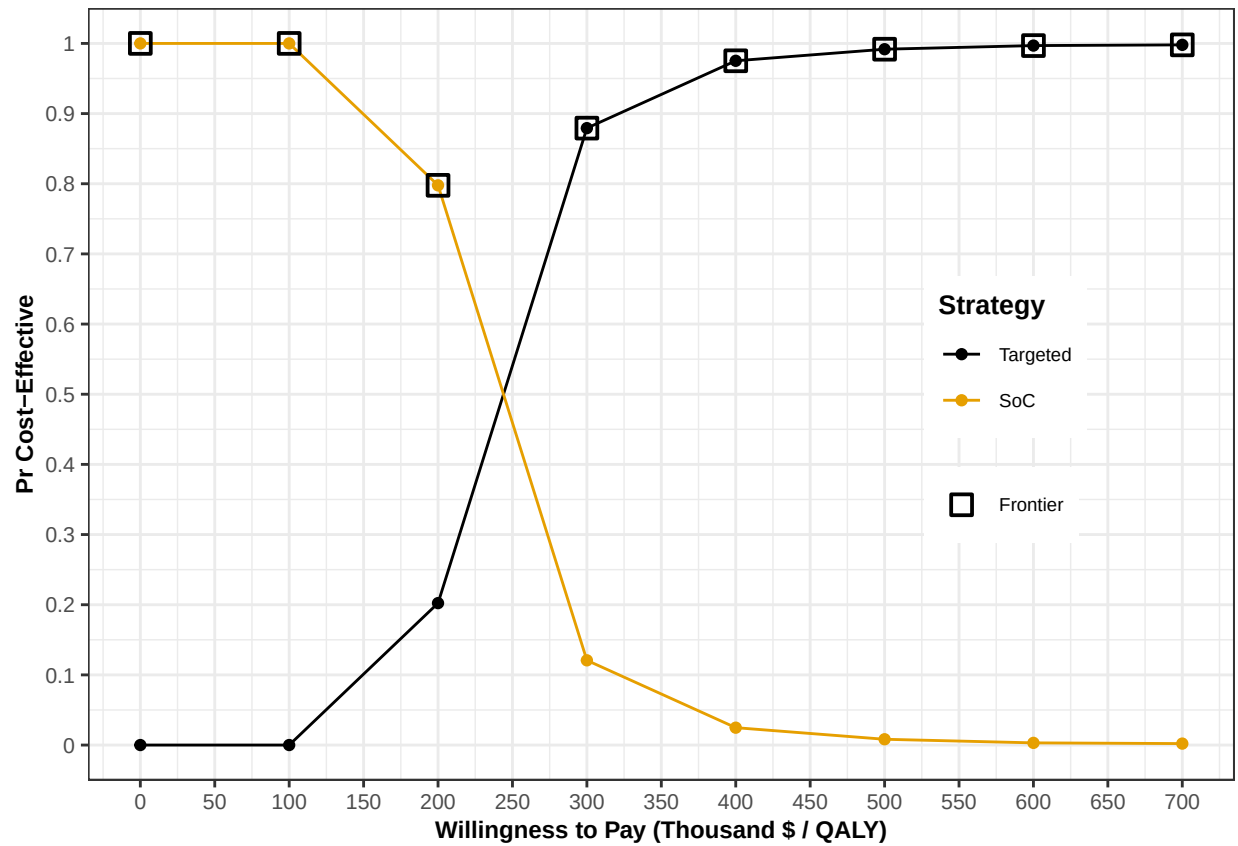
Cumulative Incidence of PLGG-related Late Effects Across Cycles





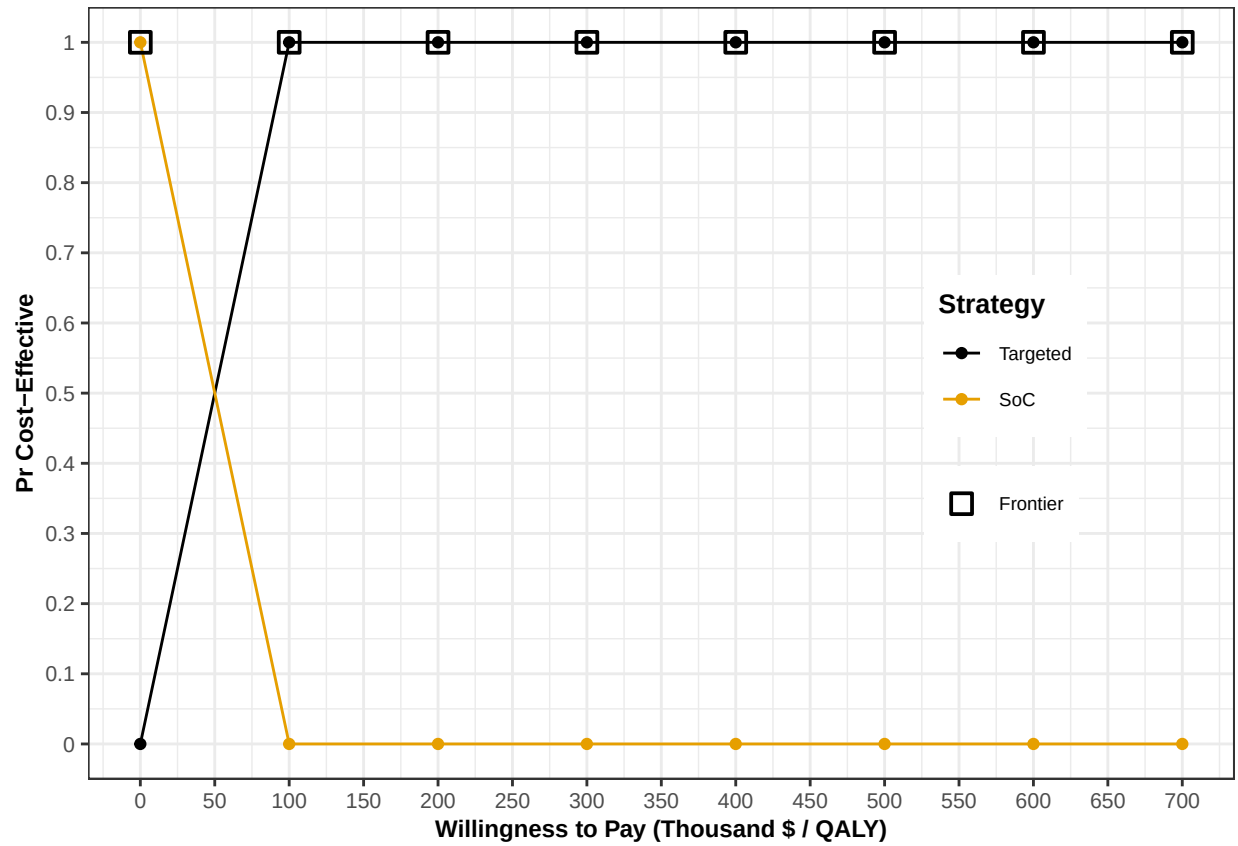
Investigator assessment, Targeted curve – exponential, SoC curve – sickkids no discount

Variable	intervention	control	Delta
Life-years	31.21 (22.55;40.8)	28.23 (18.8;39.56)	2.98 (0.02;7.86)
QALY	14.48 (10.57;18.73)	12.11 (8.23;16.89)	2.37 (1.08;4.46)
Total Cost	\$968,963 (\$718,814;\$1,361,080)	\$420,906 (\$391,640;\$456,153)	\$548,057 (\$305,050;\$934,857)
Non-Targeted PLGG related costs	\$199,778 (\$140,871;\$258,895)	\$280,185 (\$213,465;\$341,948)	\$-80,407 (\$-116,988;\$-59,892)
Late effect costs	\$80,894 (\$29,275;\$146,569)	\$108,918 (\$45,481;\$191,613)	\$-28,025 (\$-52,170;\$-7,561)
General population costs	\$48,917 (\$28,371;\$81,174)	\$31,803 (\$14,483;\$60,909)	\$17,114 (\$10,174;\$28,350)
Targeted costs	\$639,374 (\$375,448;\$1,036,833)	\$0 (\$0;\$0)	\$639,374 (\$375,448;\$1,036,833)
Radiation costs	\$0 (\$0;\$0)	\$0 (\$0;\$0)	\$0 (\$0;\$0)

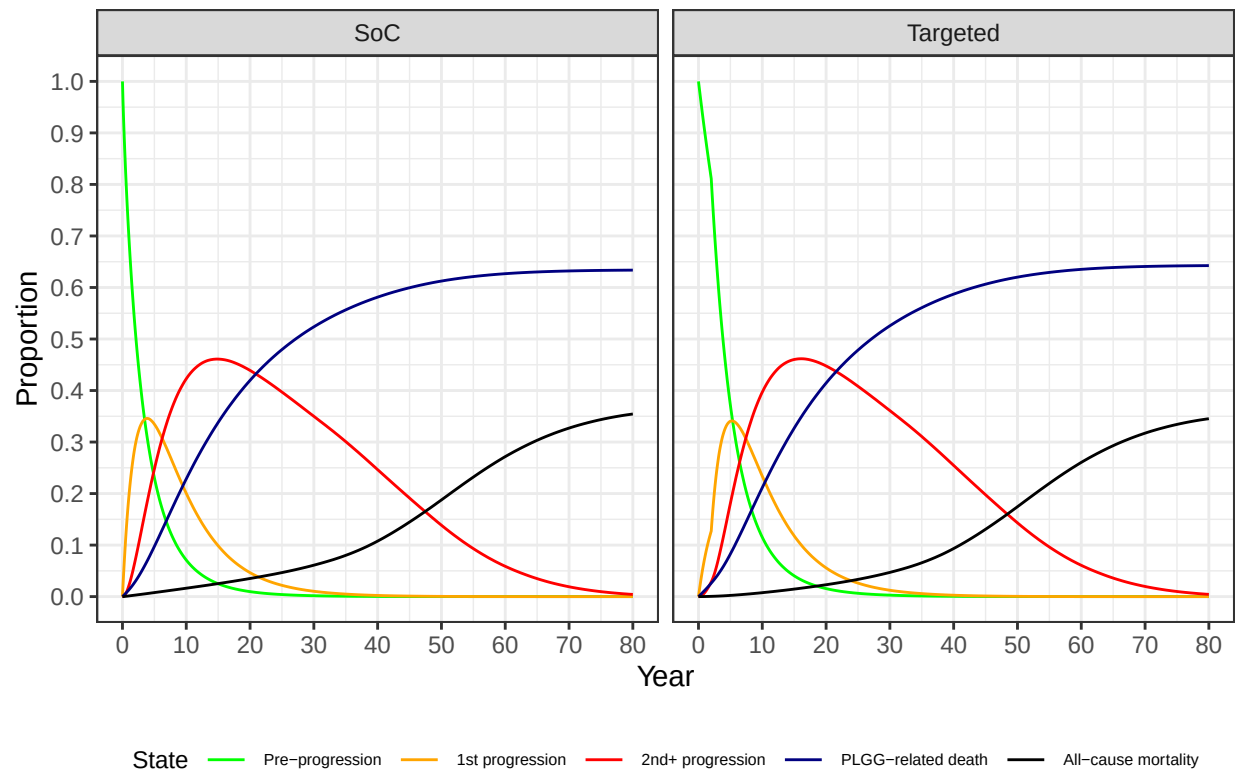


Investigator assessment, Targeted curve – exponential, SoC curve – exponential 2 year effect

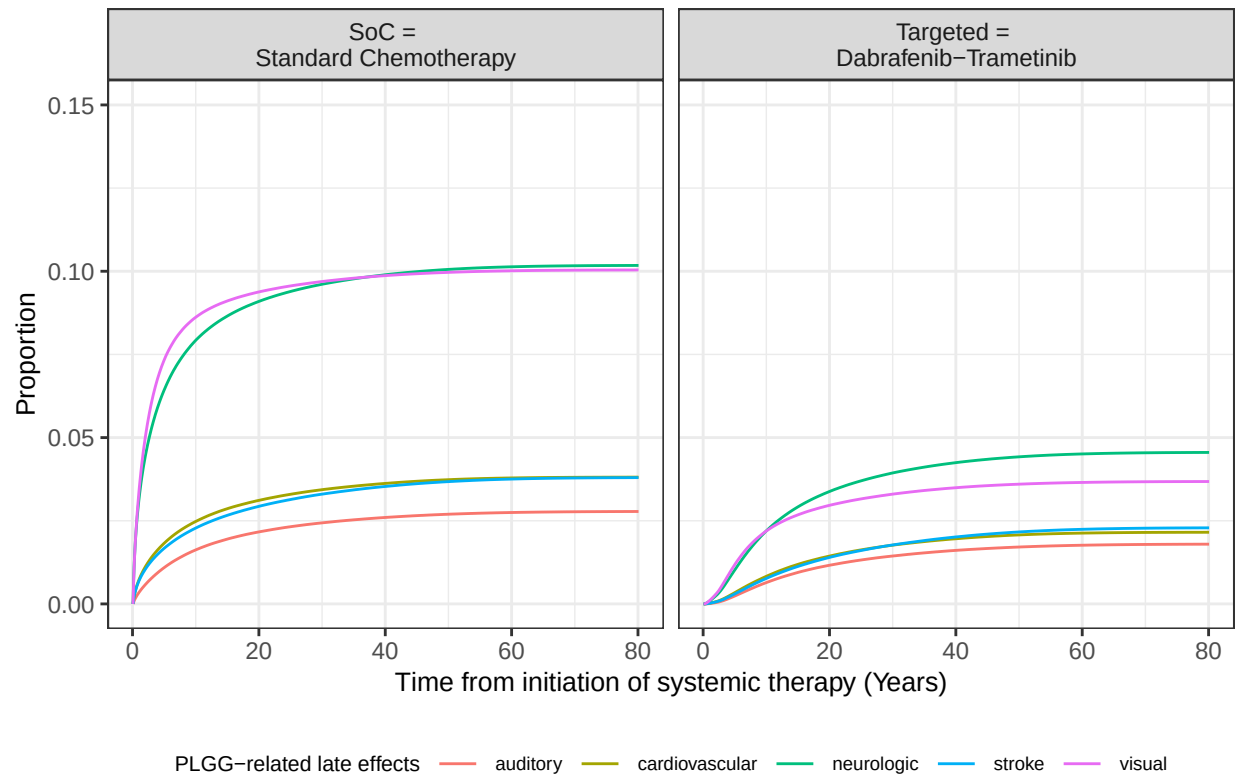
Variable	intervention	control	Delta
Life-years	29.39 (20.12;40.21)	28.62 (19.25;39.7)	0.77 (0.19;1.37)
QALY	13.49 (9.5;18.21)	12.28 (8.51;16.83)	1.21 (0.78;1.85)
Total Cost	\$469,699 (\$444,078;\$498,341)	\$415,564 (\$386,410;\$449,808)	\$54,135 (\$42,250;\$63,124)
Non-Targeted PLGG related costs	\$223,302 (\$159,219;\$285,016)	\$273,899 (\$210,008;\$334,620)	\$-50,597 (\$-56,395;\$-45,353)
Late effect costs	\$84,245 (\$27,784;\$157,926)	\$109,507 (\$47,829;\$189,429)	\$-25,262 (\$-39,061;\$-15,674)
General population costs	\$42,910 (\$23,128;\$76,135)	\$32,158 (\$15,260;\$60,585)	\$10,752 (\$7,246;\$16,872)
Targeted costs	\$119,242 (\$112,692;\$124,882)	\$0 (\$0;\$0)	\$119,242 (\$112,692;\$124,882)
Radiation costs	\$0 (\$0;\$0)	\$0 (\$0;\$0)	\$0 (\$0;\$0)

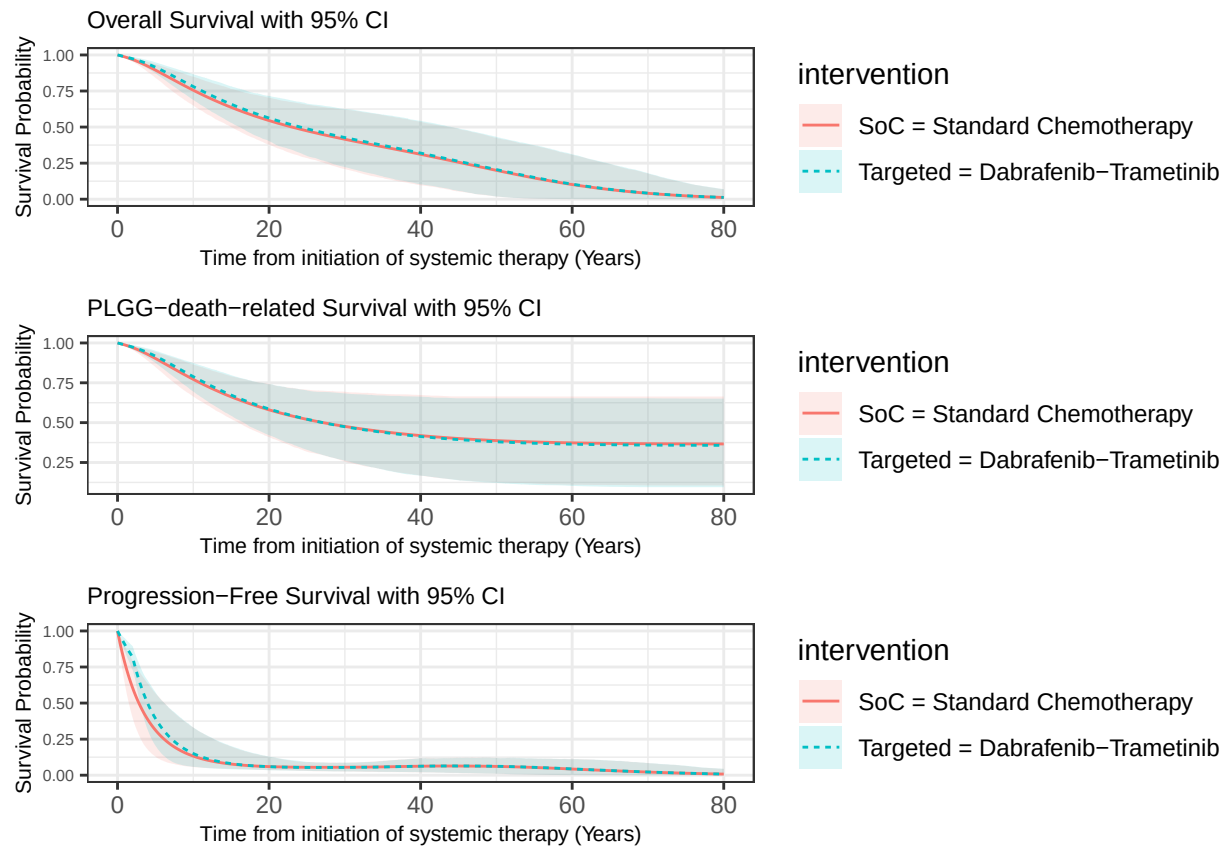


Trace Plot of PLGG States Across Cycles



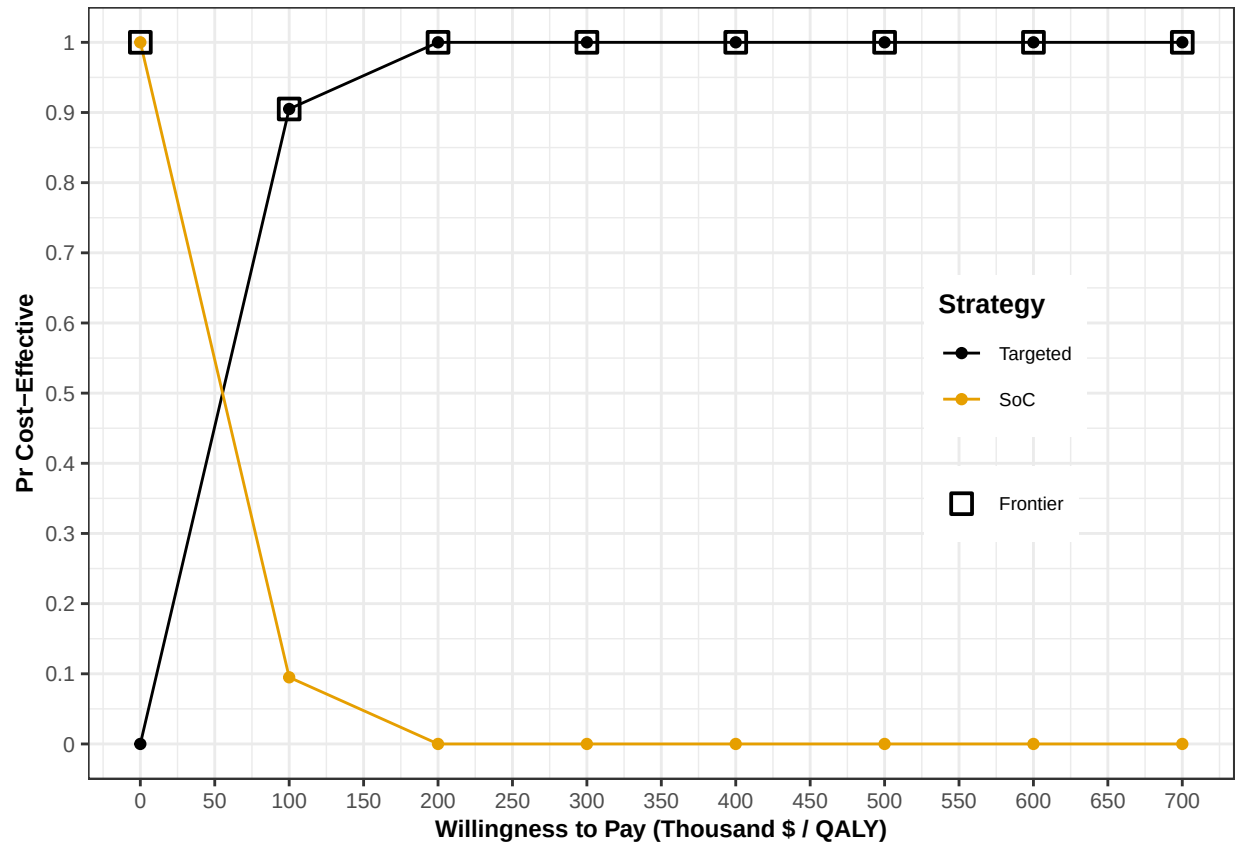
Cumulative Incidence of PLGG-related Late Effects Across Cycles



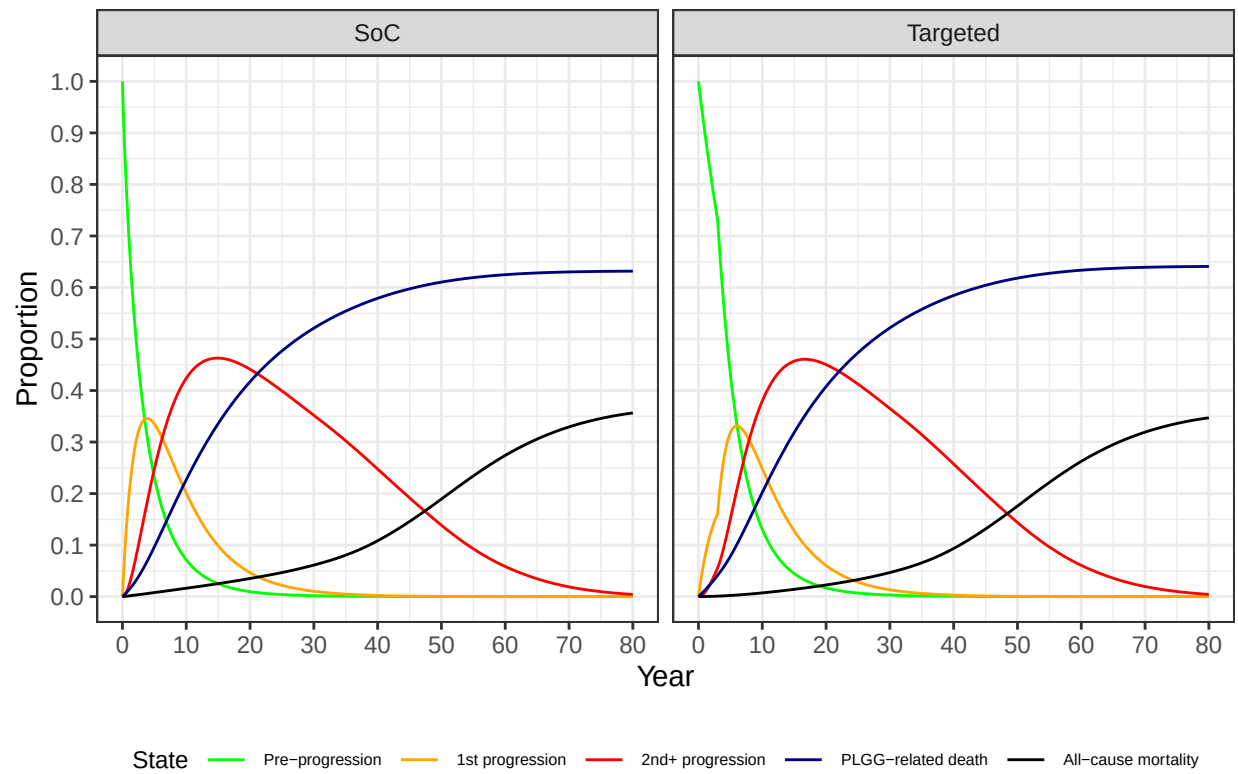


Investigator assessment, Targeted curve – exponential, SoC curve – exponential 3 year effect

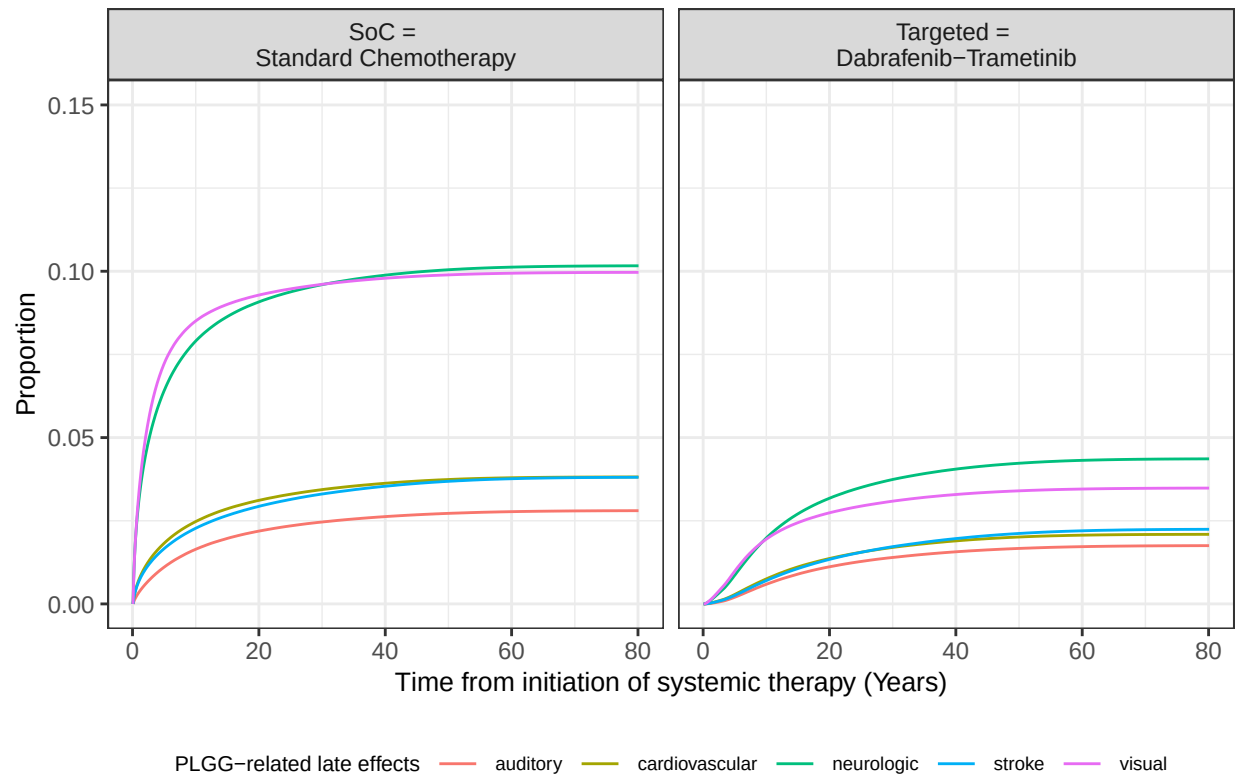
Variable	intervention	control	Delta
Life-years	29.62 (20.71;40.17)	28.7 (19.6;39.74)	0.92 (0.18;1.71)
QALY	13.63 (9.65;18.22)	12.32 (8.52;16.83)	1.31 (0.85;1.99)
Total Cost	\$514,517 (\$489,759;\$543,118)	\$415,875 (\$388,296;\$447,712)	\$98,642 (\$86,469;\$110,934)
Non-Targeted PLGG related costs	\$219,016 (\$157,149;\$279,081)	\$273,567 (\$210,291;\$333,612)	\$-54,551 (\$-61,605;\$-47,044)
Late effect costs	\$83,908 (\$27,751;\$152,959)	\$110,060 (\$46,923;\$187,692)	\$-26,152 (\$-40,926;\$-16,040)
General population costs	\$43,710 (\$24,346;\$77,390)	\$32,248 (\$15,742;\$60,851)	\$11,462 (\$7,938;\$17,658)
Targeted costs	\$167,883 (\$155,441;\$179,617)	\$0 (\$0;\$0)	\$167,883 (\$155,441;\$179,617)
Radiation costs	\$0 (\$0;\$0)	\$0 (\$0;\$0)	\$0 (\$0;\$0)

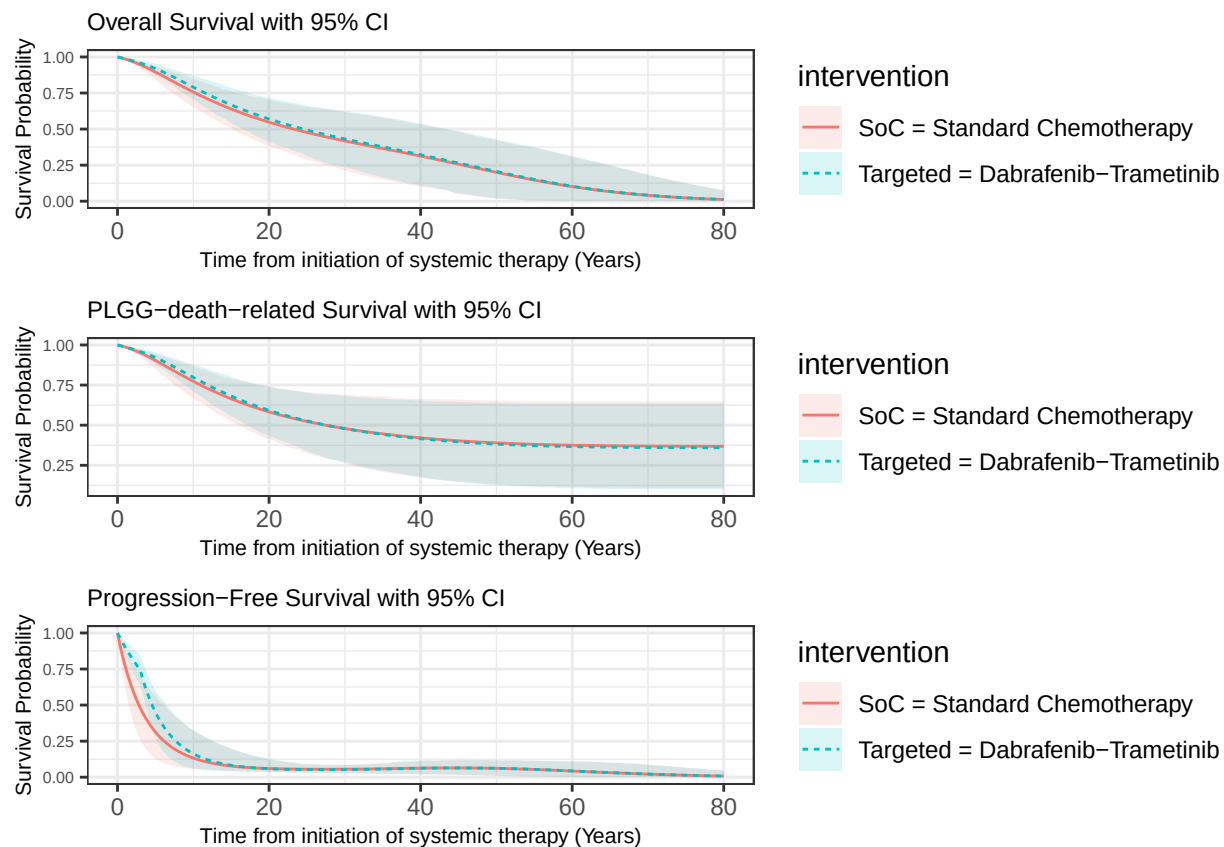


Trace Plot of PLGG States Across Cycles



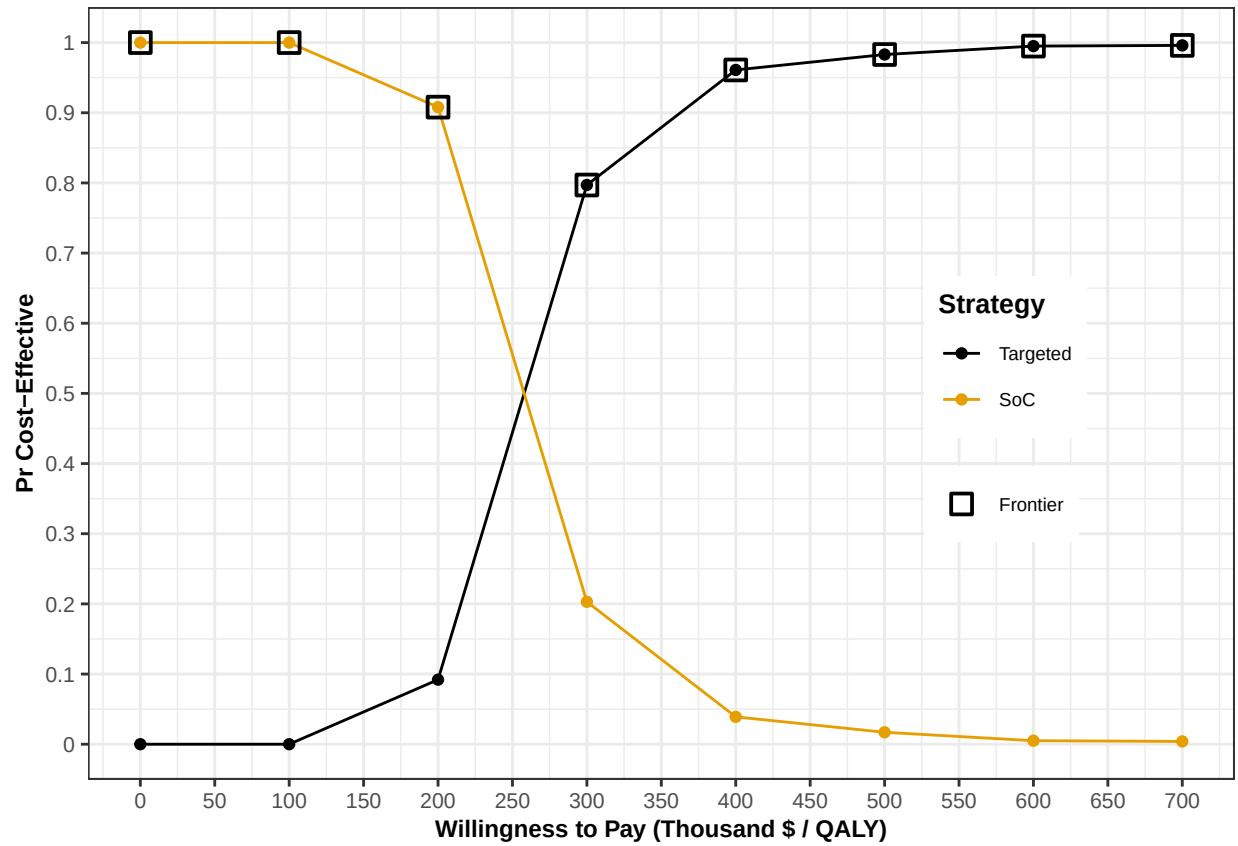
Cumulative Incidence of PLGG-related Late Effects Across Cycles



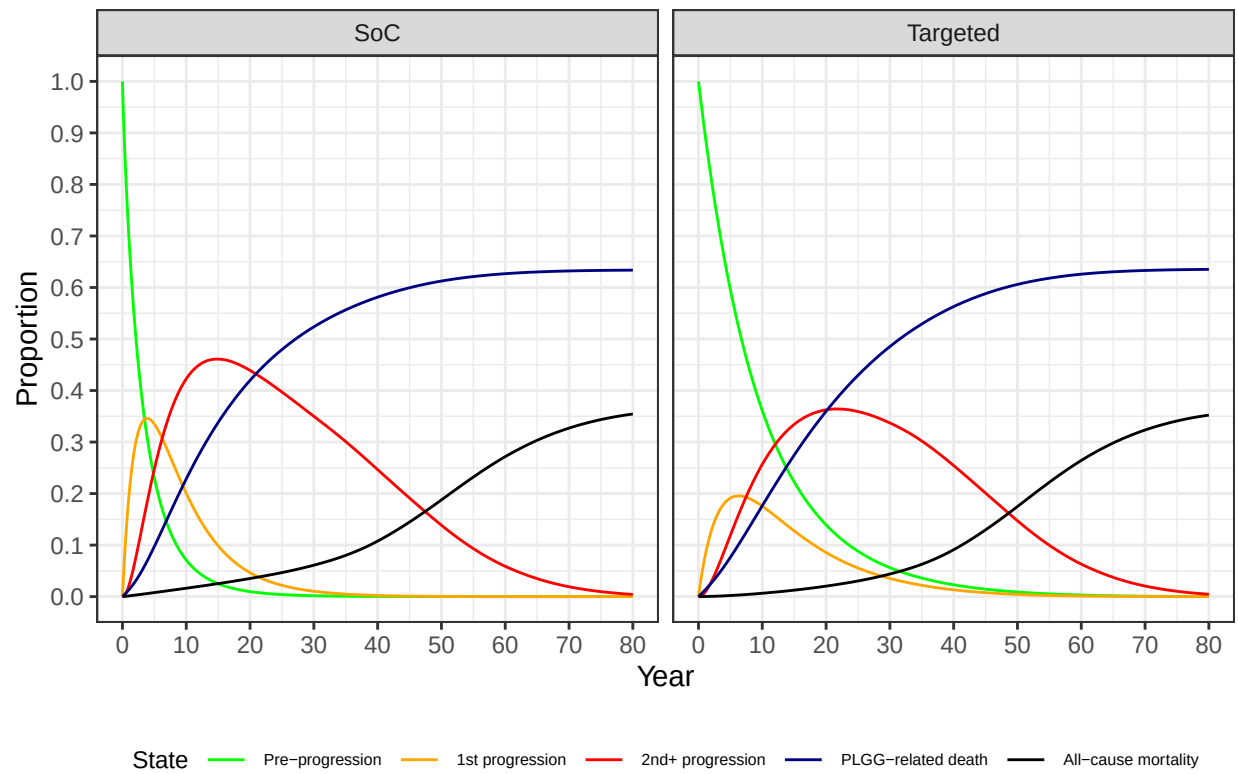


Investigator assessment, Targeted curve – exponential, SoC curve – exponential zero discounting

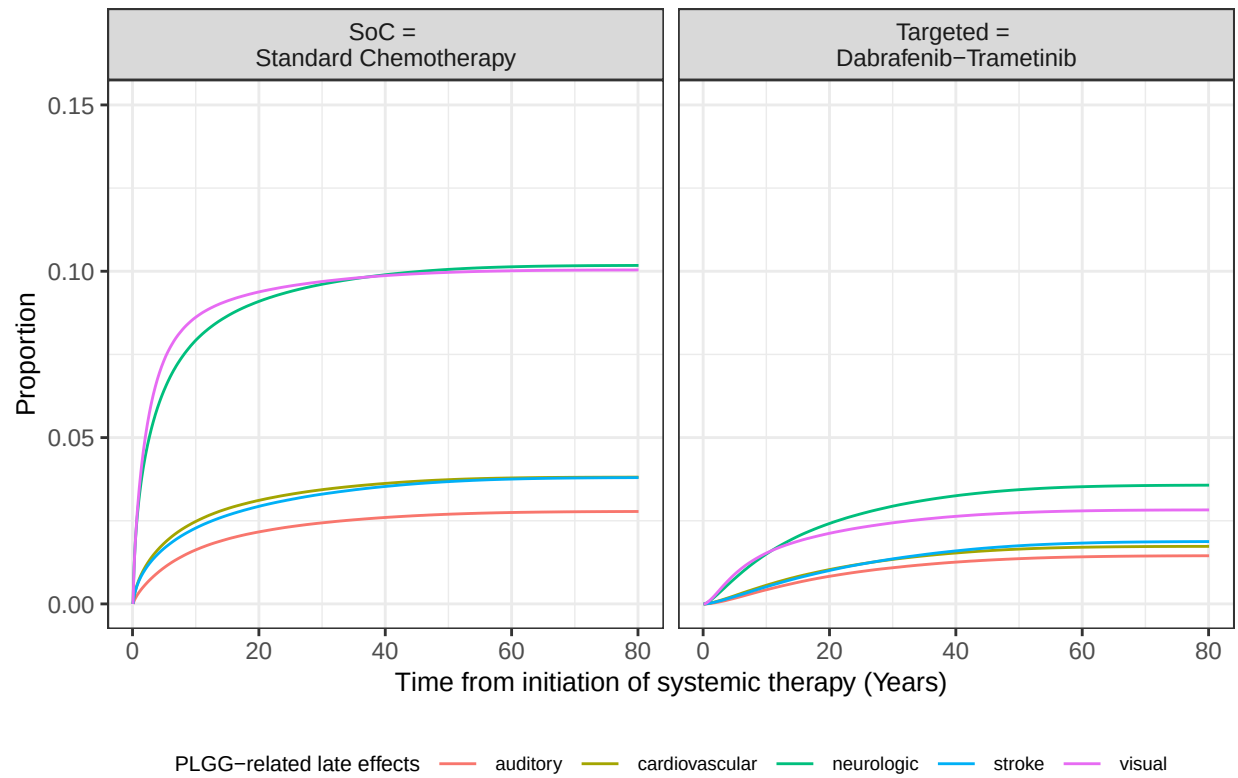
Variable	intervention	control	Delta
Life-years	31.21 (22.69;40.6)	28.62 (19.25;39.7)	2.59 (0.09;7.46)
QALY	24.86 (18.2;32.14)	21.07 (14.6;28.89)	3.79 (1.61;7.49)
Total Cost	\$1,665,162 (\$1,237,304;\$2,336,639)	\$713,034 (\$663,229;\$771,884)	\$952,128 (\$534,409;\$1,617,116)
Non-Targeted PLGG related costs	\$342,510 (\$241,992;\$444,554)	\$469,867 (\$360,376;\$574,040)	\$-127,358 (\$-186,709;\$-81,437)
Late effect costs	\$138,832 (\$50,333;\$250,620)	\$187,967 (\$82,143;\$325,183)	\$-49,135 (\$-88,061;\$-15,721)
General population costs	\$83,975 (\$49,235;\$139,027)	\$55,200 (\$26,194;\$103,992)	\$28,776 (\$16,770;\$48,179)
Targeted costs	\$1,099,846 (\$645,743;\$1,790,406)	\$0 (\$0;\$0)	\$1,099,846 (\$645,743;\$1,790,406)
Radiation costs	\$0 (\$0;\$0)	\$0 (\$0;\$0)	\$0 (\$0;\$0)

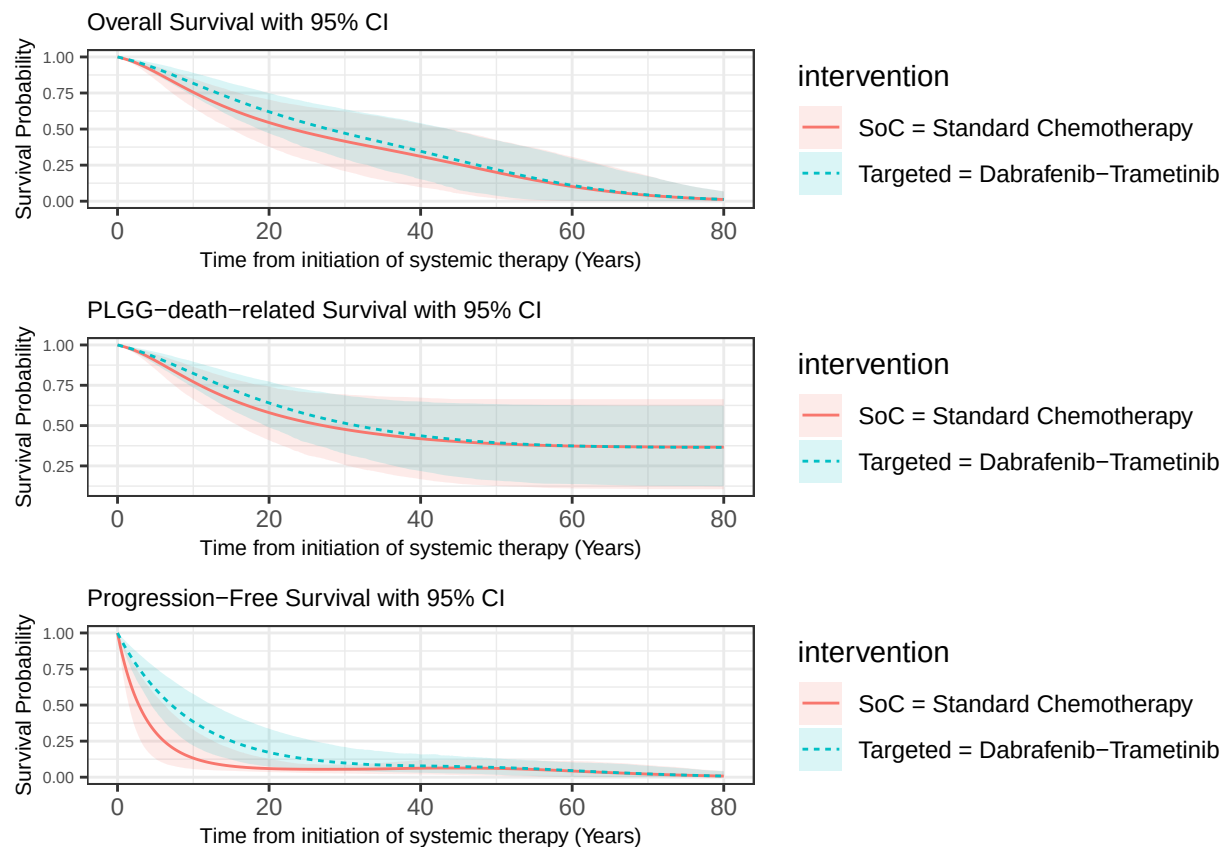


Trace Plot of PLGG States Across Cycles



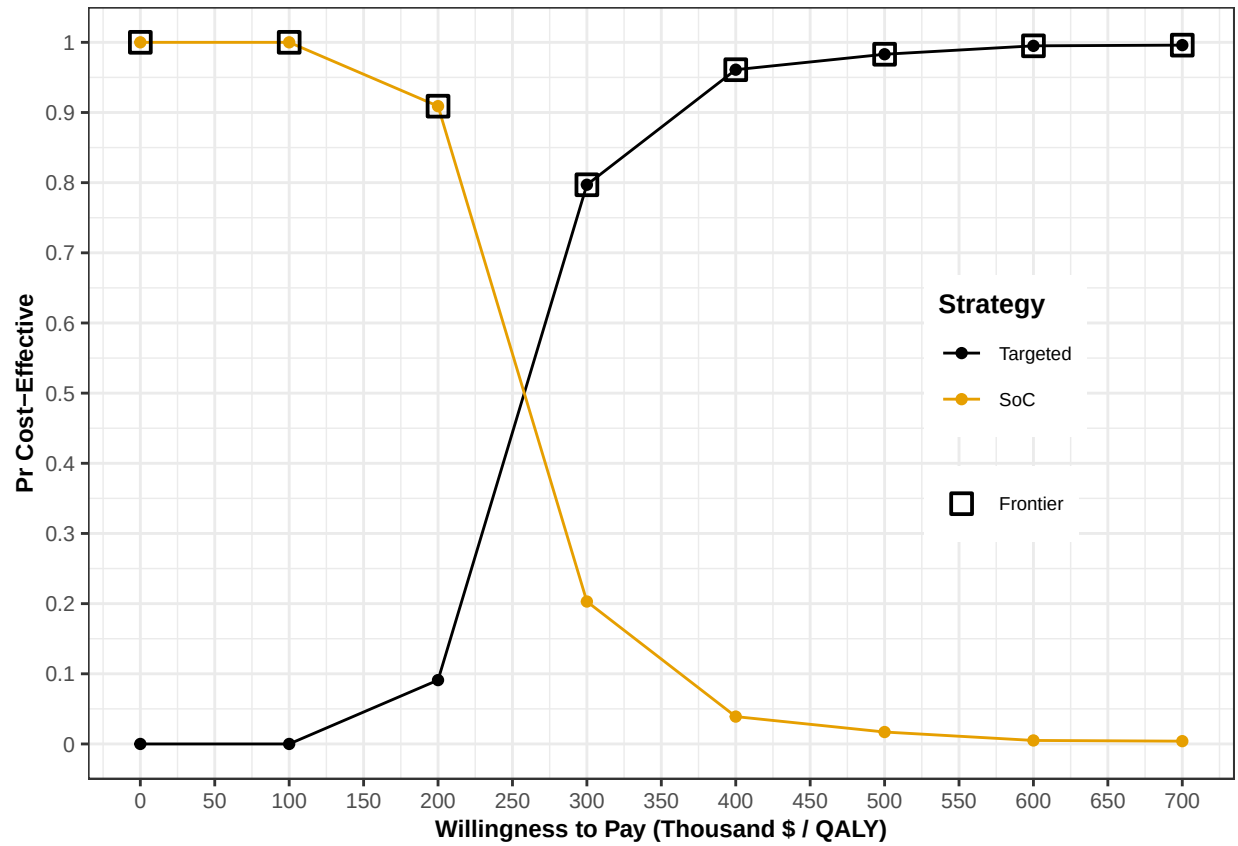
Cumulative Incidence of PLGG-related Late Effects Across Cycles



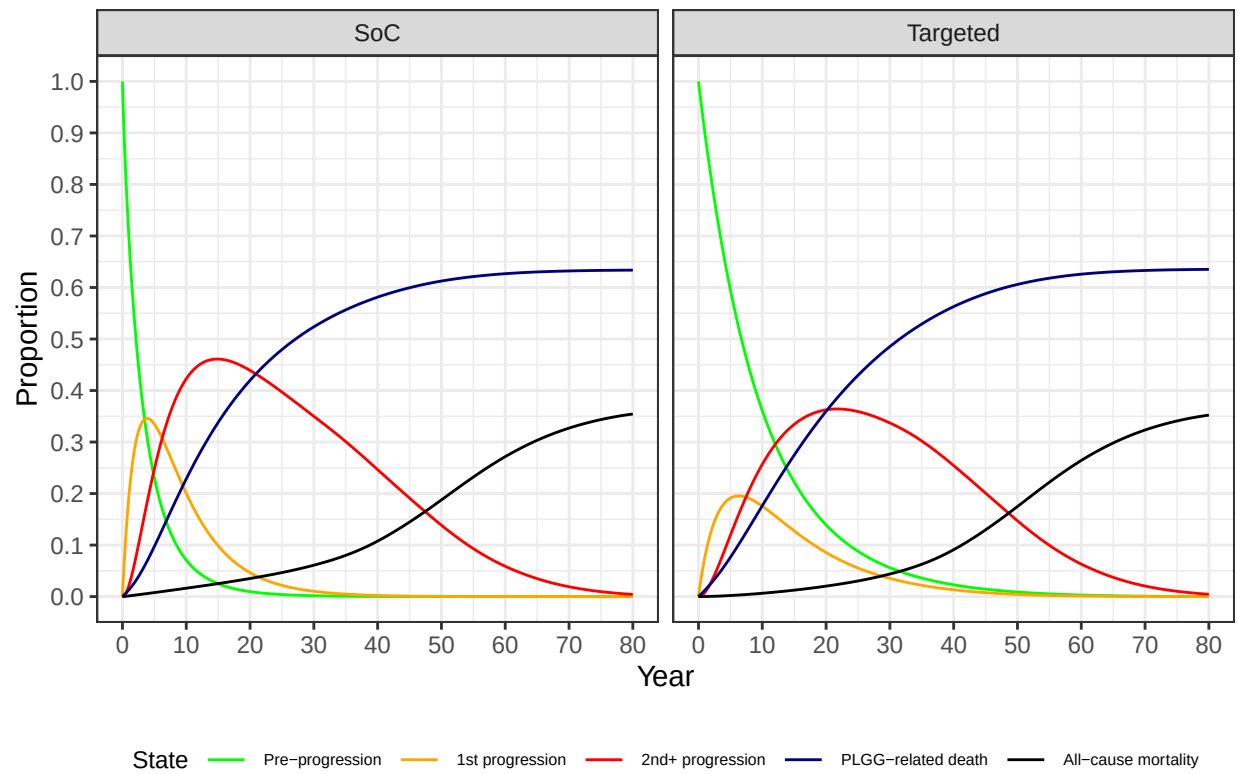


Investigator assessment, Targeted curve – exponential, SoC curve – exponential 3% discounting

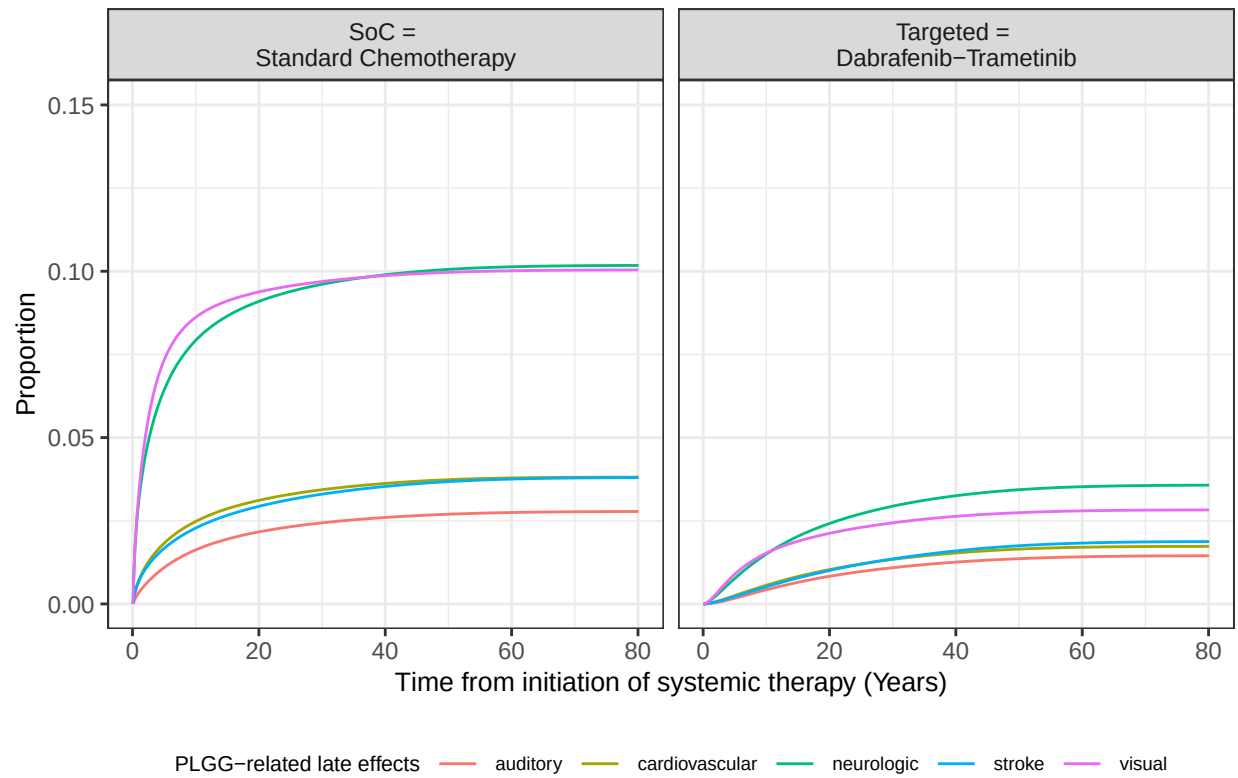
Variable	intervention	control	Delta
Life-years	31.21 (22.69;40.6)	28.62 (19.25;39.7)	2.59 (0.09;7.46)
QALY	9.43 (6.91;12.19)	8 (5.54;10.96)	1.44 (0.61;2.84)
Total Cost	\$632,055 (\$469,642;\$886,795)	\$270,725 (\$251,685;\$292,993)	\$361,331 (\$202,916;\$613,730)
Non-Targeted PLGG related costs	\$130,176 (\$92,083;\$168,605)	\$178,467 (\$136,784;\$217,981)	\$-48,290 (\$-70,782;\$-30,869)
Late effect costs	\$52,676 (\$19,130;\$95,042)	\$71,315 (\$31,133;\$123,354)	\$-18,639 (\$-33,369;\$-5,937)
General population costs	\$31,860 (\$18,679;\$52,749)	\$20,943 (\$9,938;\$39,455)	\$10,917 (\$6,362;\$18,275)
Targeted costs	\$417,343 (\$245,065;\$679,327)	\$0 (\$0;\$0)	\$417,343 (\$245,065;\$679,327)
Radiation costs	\$0 (\$0;\$0)	\$0 (\$0;\$0)	\$0 (\$0;\$0)

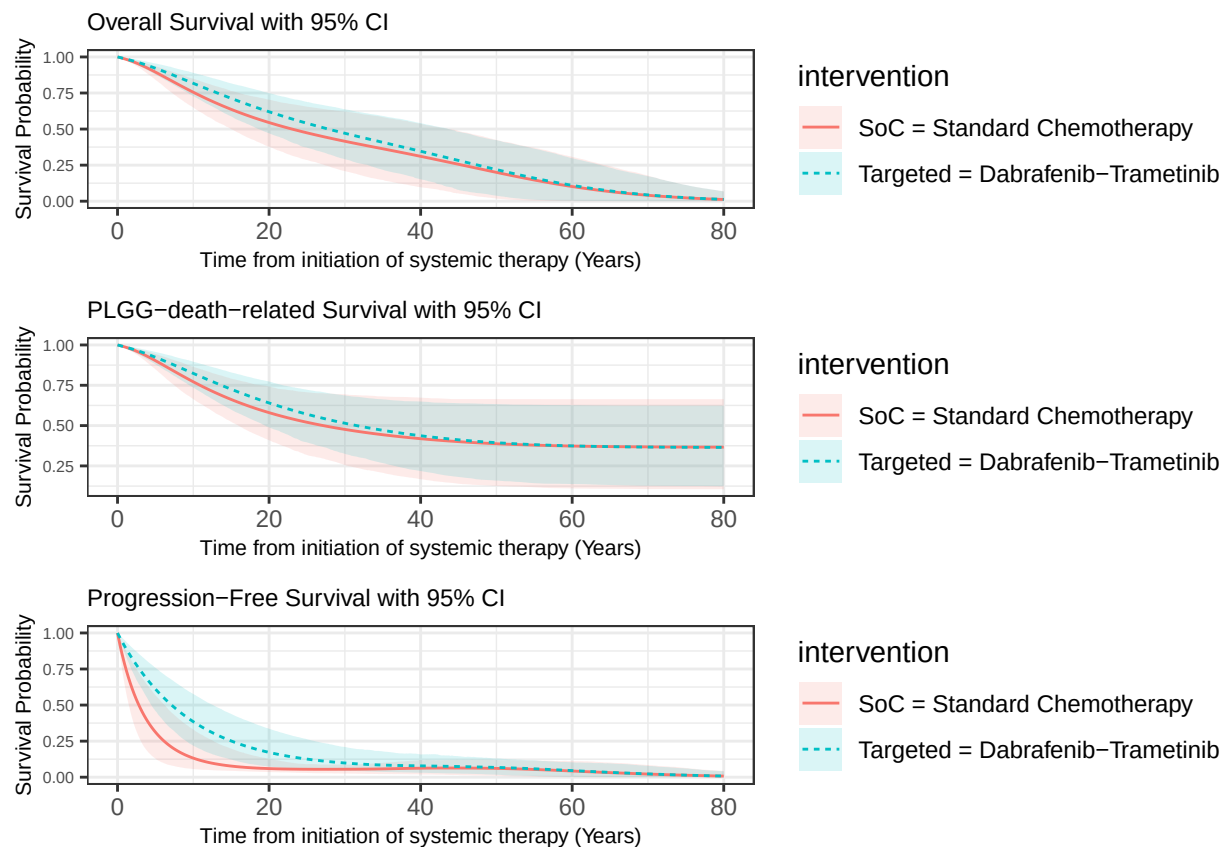


Trace Plot of PLGG States Across Cycles



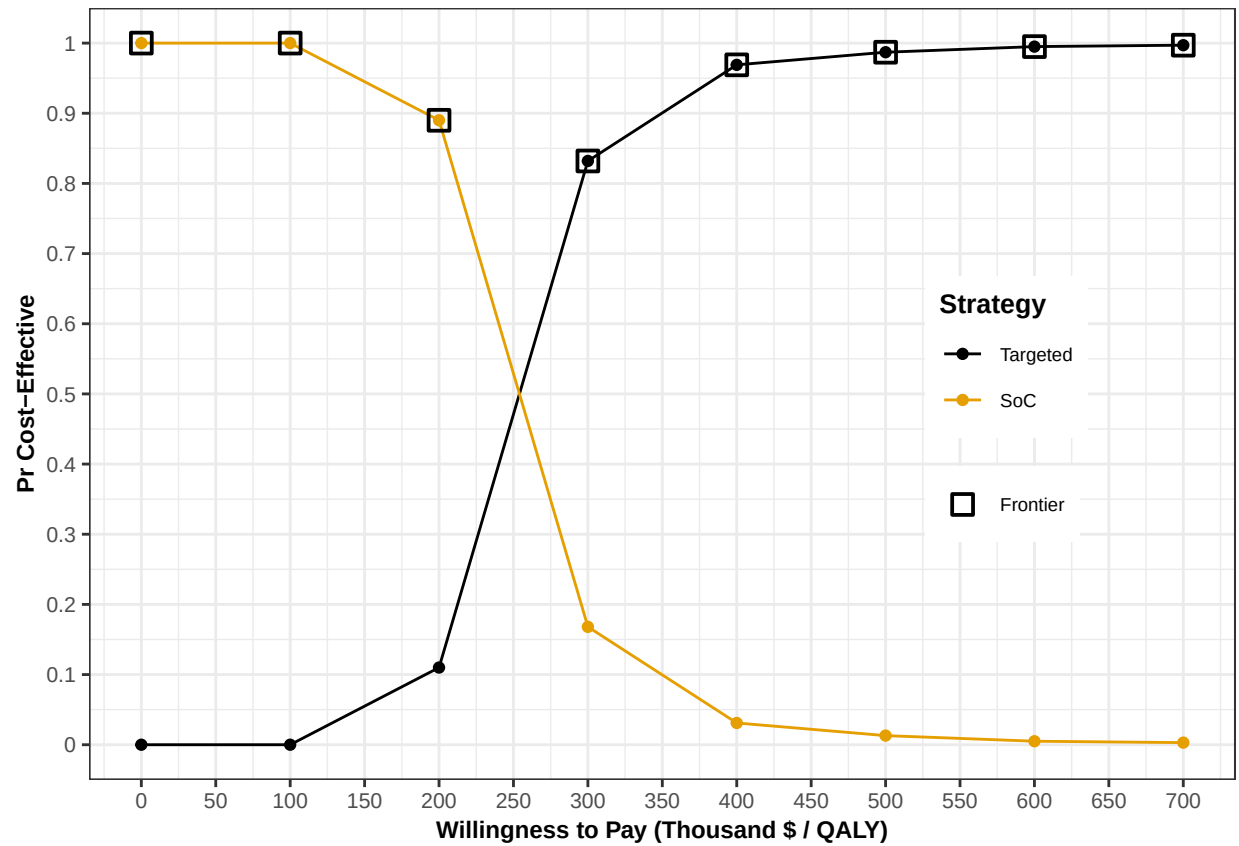
Cumulative Incidence of PLGG-related Late Effects Across Cycles



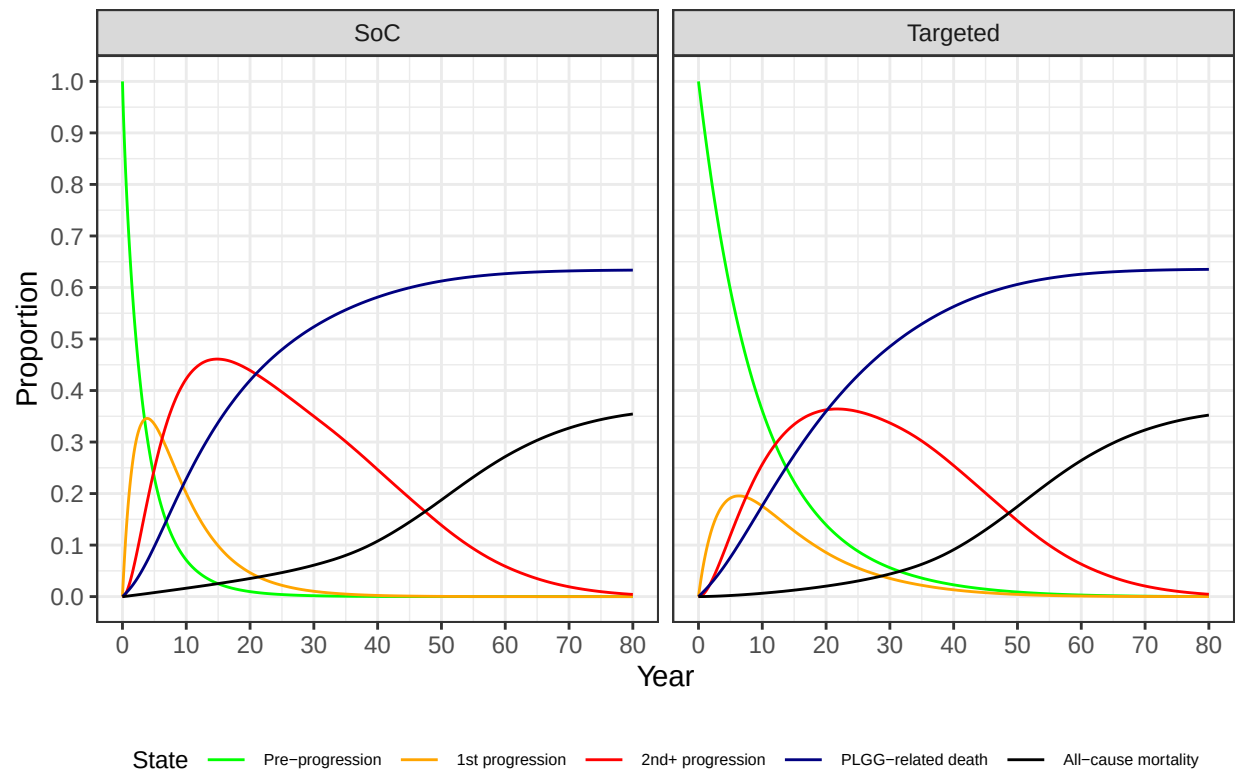


Investigator assessment, Targeted curve – exponential, SoC curve – exponential lifelong costs

Variable	intervention	control	Delta
Life-years	31.21 (22.69;40.6)	28.62 (19.25;39.7)	2.59 (0.09;7.46)
QALY	14.48 (10.6;18.72)	12.28 (8.51;16.83)	2.21 (0.94;4.36)
Total Cost	\$1,161,684 (\$902,379;\$1,567,427)	\$624,592 (\$516,450;\$750,882)	\$537,092 (\$307,410;\$907,967)
Non-Targeted PLGG related costs	\$425,944 (\$354,337;\$497,839)	\$514,938 (\$450,039;\$595,269)	\$-88,993 (\$-144,964;\$-50,771)
Late effect costs	\$80,883 (\$29,349;\$145,978)	\$109,507 (\$47,829;\$189,429)	\$-28,623 (\$-51,273;\$-9,141)
General population costs	\$14,053 (\$7,905;\$25,139)	\$147 (\$0;\$832)	\$13,905 (\$7,474;\$25,070)
Targeted costs	\$640,803 (\$376,259;\$1,043,104)	\$0 (\$0;\$0)	\$640,803 (\$376,259;\$1,043,104)
Radiation costs	\$0 (\$0;\$0)	\$0 (\$0;\$0)	\$0 (\$0;\$0)



Trace Plot of PLGG States Across Cycles



Cumulative Incidence of PLGG-related Late Effects Across Cycles

